COMSATS University Islamabad, Pakistan



AI-Assisted Plasma Process Experimentation & Digital Twin Analytics Platform

Software Requirements Specification (SRS)

Version 1.1

By

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Revision History

The table below records every released version of this Software Requirements Specification. A new row is added whenever the document is changed after review, so evaluators can see what was changed, when and why.

Version	Date	Description of change	Author(s)
1.0	25-Oct-2026	Initial SRS for the revised three-stage scope (low-pressure argon plasma with OES and Langmuir probe; underwater oxygen plasma for disinfection). Supersedes the requirements implied by Scope Document v2.0, following the supervisor's redirection of the project (Section 1.6.2).	M. Rayyan Taimoor Khan, Sadeem Ur Rehman
1.1	29-Oct-2026	AI made the second core module (no longer optional): Core Module 1 Digital Twin and Core Module 2 AI. Added objectives BO-6 to BO-9, sub-modules 2.1–2.5, mockups M13–M16 and their requirements, use cases UC-24 to UC-28, backend events E23–E27, AI model-quality requirements AIQ-1 to AIQ-6 and constraint CON-12; updated the architecture, traceability and Plagiarism Report chapter. Event requirements renumbered FR13.x → FR17.x.	M. Rayyan Taimoor Khan, Sadeem Ur Rehman

Abstract

Plasma diagnostics such as optical emission spectroscopy (OES) and Langmuir probes are central to plasma physics teaching and research, yet students can rarely practise them because laboratory plasma systems are expensive, shared and slow to operate. At present, students either wait for limited time on real equipment or use separate specialist programs, each of which covers only one piece of the problem, such as electron kinetics, spectrum synthesis or probe-curve fitting. Two challenges follow: a student cannot see how a change in RF power, pressure or gas flow propagates through the plasma state into the measured spectrum and probe curve, and there is no simple way to check those diagnostics against each other or against measured data. The same gap exists for underwater plasma disinfection, where discharge settings are rarely connected quantitatively to the reactive species produced and the resulting bacterial log reduction. No single, low-cost, physics-based tool links discharge settings, plasma state, synthetic diagnostics and validation for both a low-pressure and an underwater plasma. This project will build a software digital twin of a 13.56 MHz capacitively coupled argon plasma (10–100 W, 0.1–0.5 mbar, 100–500 sccm, with optional N_2 and O_2) whose electron temperature, electron density and gas temperature come from particle and power balance equations. It will generate and analyse virtual OES spectra (Boltzmann plot, excitation and gas temperature) and virtual Langmuir probe curves (OML theory, Druyvesteyn EEDF), and it will model an underwater pulsed-DC oxygen bubble discharge whose reactive species drive a Hom-model prediction of *E. coli* inactivation. All diagnostics will be cross-checked against the model and against published or measured data within a fixed ±30 % tolerance. A second core module applies AI to the twin: it infers plasma parameters from spectra with stated uncertainty, detects runs that break the expected physics, explains the twin through a fast surrogate, and recommends settings whose outcome is re-simulated by the twin. The result is a free, repeatable virtual plasma laboratory that lets students and researchers rehearse experiments, interpret diagnostics and validate measurements before and after using the real laboratory.

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Abbreviations

The table below lists the abbreviations and symbols used in this document.

Abbreviation / Symbol	Meaning
0D / 1D	Zero-dimensional (volume-averaged) / one-dimensional (axial) model
AIQ	AI model-quality requirement
BO	Business Objective
BR	Business Rule
CCP	Capacitively Coupled Plasma
CFU	Colony-Forming Unit
COD	Chemical Oxygen Demand
CR model	Collisional-Radiative model
CSV	Comma-Separated Values
DC	Direct Current
EEDF / EEPF	Electron Energy Distribution / Probability Function
FE	Feature
FR	Functional Requirement
FWHM	Full Width at Half Maximum
I-V	Current–voltage characteristic of a Langmuir probe
LXCat	Plasma Data Exchange Project (electron cross-section database)
M1 … M16	Mockup identifiers
MAE	Mean Absolute Error
ML	Machine Learning
NFR	Non-Functional Requirement
NIST ASD	NIST Atomic Spectra Database
$\mathbf{n_e}$	Electron density (m^{-3})
OES	Optical Emission Spectroscopy
OML	Orbital Motion Limited (probe theory)
ORP	Oxidation–Reduction Potential
RF	Radio Frequency
sccm	Standard cubic centimetres per minute
SHAP	SHapley Additive exPlanations
SRS	Software Requirements Specification
$\mathbf{T_e}$	Electron temperature (eV)
$\mathbf{T_{exc}}$	Excitation temperature (eV)
$\mathbf{T_{gas}}$	Gas (neutral) temperature (K)
UC	Use Case
$\mathbf{V_f}$ / $\mathbf{V_p}$	Floating potential / plasma potential (V)
$\boldsymbol{\lambda_D}$	Electron Debye length
1 lac	100 000 (used for colony counts)

Chapter 1: Introduction and Problem Definition

This chapter introduces the Plasma Digital Twin Platform. It states the vision, the problem and the proposed solution, sets measurable business objectives, defines the scope and its limits, records how the scope was revised after supervisor review, lists the modules and their features, reviews related systems and literature, and closes with the tools, the contribution of the project and its relevance to the BSCS course modules.

1.1 Overview of the Project

The AI-Assisted Plasma Process Experimentation & Digital Twin Analytics Platform is a desktop, browser-based virtual plasma laboratory for plasma physics students and researchers. It reproduces, in software, the three experimental stages defined by the supervisor: **Stage 1**, a low-pressure argon plasma generated in a 13.56 MHz capacitively coupled (CCP) chamber and diagnosed by optical emission spectroscopy (OES); **Stage 2**, an underwater high-pressure oxygen plasma generated inside gas bubbles by a pulsed DC supply and used to disinfect municipal waste water; and **Stage 3**, Langmuir probe diagnostics, which are applied to the Stage 1 plasma only. For each stage the user sets the same controls that exist on the real equipment (RF power, pressure, gas flow and mixture, or pulse voltage, frequency and electrode gap), the platform solves physics-based models for the plasma state, and it then produces the signals a real instrument would record: an emission spectrum, a probe current–voltage (I-V) curve, or a bacterial survival curve. These signals are analysed with the same methods used in the laboratory, compared with each other and with the model, and validated against published or measured data. The platform has two core modules: **Core Module 1, the Digital Twin**, which holds all of this physics, and **Core Module 2, the AI Module**, which learns from data generated by the twin to infer plasma parameters from spectra, detect physics-relationship anomalies, explain the twin through a fast surrogate, and recommend operating settings that are verified by re-running the twin.

1.2 Vision Statement

For plasma physics students and researchers at COMSATS University Islamabad **who** need to practise and understand plasma diagnostics without waiting for scarce laboratory time, **the** Plasma Digital Twin Platform **is** a physics-based virtual plasma laboratory **that** lets them configure a low-pressure argon plasma and an underwater oxygen plasma, obtain the resulting plasma state, generate and analyse realistic OES spectra and Langmuir probe curves, predict disinfection performance, and validate every result against published and measured data. **Unlike** specialist commercial simulators and single-purpose analysis programs, which are expensive or cover only one step of the chain, **our product** connects discharge settings, plasma physics, synthetic diagnostics and validation in one free, transparent and repeatable environment in which every number shown can be traced back to a published equation or data source.

1.3 Problem Statement

Plasma diagnostics are learned by doing, but doing is expensive. A low-pressure RF plasma system needs a vacuum chamber, pumps, gas controllers, an RF generator with a matching network, a spectrometer and a probe system, and the laboratory is shared between many users. A student therefore gets only a few hours of operating time, often without being able to repeat a run or explore the full range of power, pressure and gas flow. Interpreting the results is also hard. The emission spectrum depends on the electron temperature, the electron density and the electron energy distribution through a chain of atomic processes, and a probe curve is distorted by the RF sheath oscillation; students who see only the final spectrum or curve cannot tell which physical cause produced which feature. Existing software does not close this gap. General multiphysics packages are costly and require expertise in finite-element modelling, while free tools each address one step (electron kinetics, spectrum synthesis or probe fitting) and must be stitched together by hand in spreadsheets. The same problem exists for underwater plasma disinfection, where the link from pulse voltage, frequency and gas flow to reactive species and finally to bacterial log reduction is usually studied only by trial and error. Finally, when measured data become available, there is no consistent, tolerance-based way to check them against theory and against the other diagnostic, so errors in either the experiment or the analysis can go unnoticed.

1.4 Problem Solution

The proposed system solves the problem with a digital twin for each plasma and a virtual instrument for each diagnostic, all connected through one plasma state. For Stage 1, a volume-averaged (0D) global model solves the particle balance for the electron temperature and the power balance for the electron density [1, 2], extended with gas flow, gas temperature, $Ar/N_2/O_2$ chemistry, a Druyvesteyn electron energy distribution and a 1D axial profile. The virtual OES uses a collisional-radiative (CR) model of the argon 1s and 2p levels [3] and NIST line data [4] to synthesise a spectrum, which is then analysed with a Boltzmann plot, line-ratio methods and rotational or Doppler gas-temperature methods. The virtual Langmuir probe computes an RF-distorted I-V curve and analyses it with OML theory and the Druyvesteyn method [5, 6]. For Stage 2, a gas-phase global model of the oxygen–water-vapour bubble discharge feeds an empirical transfer of reactive species into the water and a Hom disinfection model [7]. A comparison screen places model, OES and probe values side by side with a fixed ±30 % tolerance, and a researcher can import measured OES and probe files for validation only. Every run can be saved, compared and exported. The second core module, the AI module, learns from data generated by the twin: tree-ensemble models infer plasma parameters from a spectrum with conformal prediction intervals, an Isolation Forest flags runs whose diagnostics break the expected physics relationship, a surrogate with SHAP explains which inputs drive each output, and a recommendation engine proposes settings for a goal and re-simulates each one through the twin before showing it.

1.5 Objectives of the Proposed System

The business objectives below are measurable so that the final product can be evaluated against them. Each objective is traced to modules, use cases and requirements in Appendix B.

BO-1: Provide a virtual low-pressure argon CCP experiment that covers the whole supervisor-defined operating envelope (10–100 W, 0.1–0.5 mbar, Ar 100–500 sccm, N_2 and O_2 admixtures) and whose computed T_e, n_e and T_{gas} agree with published reference values within ±30 % at no fewer than three reference operating points.

BO-2: Allow a student to configure a Stage 1 run and obtain a synthetic OES spectrum and a Langmuir probe analysis for it in under 5 minutes, with every derived quantity (T_e, n_e, T_{gas}, T_{exc}) reported with its uncertainty and method.

BO-3: Demonstrate diagnostic consistency: OES- and probe-derived values shall agree with the model within ±30 % for at least 90 % of a pre-registered grid of Stage 1 test configurations.

BO-4: Predict the *E. coli* log reduction of the underwater oxygen discharge within ±30 % of published reference data at no fewer than one reference operating point, together with the D-value and energy yield.

BO-5: Allow a researcher to import a measured OES spectrum (.txt) or probe I-V curve (.csv) and obtain a validation report in under 10 minutes, without measured data ever being used to tune model parameters.

BO-6: Infer T_e, n_e and T_{gas} from an emission spectrum with an AI model whose error is significantly lower than a ridge-regression baseline (paired bootstrap, $p < 0.05$) and whose 90 % prediction intervals reach at least 85 % coverage on held-out synthetic spectra.

BO-7: Detect physics-relationship faults injected by the twin (probe contamination, wavelength shift, air leak, power-meter error, RF compensation failure) with at least 90 % recall at no more than 5 % false alarms, significantly better than a range-check baseline.

BO-8: Provide a surrogate of the twin with $R^2 \geq 0.98$ and median relative error ≤ 5 % on held-out twin runs, at least 100 times faster than the twin, with SHAP explanations of its outputs.

BO-9: Recommend operating settings for a user goal such that 100 % of recommendations are re-simulated by the twin and the twin confirms the goal is met for at least 90 % of a benchmark set of goals.

1.6 Scope

The scope of the project is a single-user, locally installed software platform that simulates and analyses two laboratory plasmas and two diagnostic techniques. The platform simulates a low-pressure argon plasma in a 13.56 MHz capacitively coupled chamber of 40 cm by 40 cm with

electrodes 14 cm in diameter and 4.5 cm apart. The user controls RF power from 10 to 100 W, pressure from 0.1 to 0.5 mbar, argon flow from 100 to 500 sccm and the gas mixture, where nitrogen and oxygen are each added as 5 % of the argon flow by default. The Stage 1 model is a volume-averaged global model that solves particle and power balance for the electron temperature and electron density and an energy balance for the gas temperature, with a Druyvesteyn electron energy distribution and an optional one-dimensional axial profile. A virtual OES module produces the emission spectrum of this plasma from a collisional-radiative model, adds the effect of the spectrometer resolution and noise, and analyses the spectrum with a Boltzmann plot, the excitation temperature, the excited-level population distribution and the gas temperature from Doppler broadening or the nitrogen rotational band. A virtual Langmuir probe produces the current–voltage curve of a cylindrical tungsten probe in the same plasma, including RF sheath distortion, and analyses it with orbital-motion-limited theory to obtain the floating and plasma potentials, electron temperature, electron density, Debye length and the electron energy distribution. A comparison function shows model, OES and probe values side by side and marks each as inside or outside a fixed ±30 % tolerance. The platform also simulates an underwater oxygen plasma generated inside bubbles by a pulsed DC supply of up to 10 kV at 5 to 10 kHz and 100 W average power across a pin-to-pin gap of 1 to 3 mm at 1 atm. For this plasma it computes the bubble plasma state with a Maxwellian distribution, the concentrations of ozone, hydrogen peroxide and hydroxyl radicals in the water, the change in pH, oxidation–reduction potential, conductivity and temperature, and the survival of *E. coli* and other listed organisms with the Hom model. Its virtual OES shows the OH, H-alpha and atomic oxygen lines and estimates the electron density from H-alpha Stark broadening. In researcher mode, the user can import measured spectra and probe curves, attach the required metadata, and validate the models against these files and against published benchmarks, while measured data stay non-citable unless permission is given. All runs can be saved, reopened, compared and exported as tables, figures or a PDF report. The AI module, the second core module, is trained only on synthetic data generated by the twin. It infers electron temperature, electron density and gas temperature from a spectrum with prediction intervals, flags runs and measured files whose diagnostics break the expected physics relationship, offers a fast surrogate with SHAP explanations, and recommends settings for a user goal, each of which is re-simulated by the twin. Control of real laboratory hardware, other gases, multi-dimensional fluid or particle simulations and multi-user accounts are outside the scope.

1.6.1 Limitations/Constraints

LI-1: The platform is software-only. It does not control or acquire data directly from laboratory hardware; measured data enter only as exported files.

LI-2: Stage 1 is limited to argon with nitrogen and/or oxygen admixtures of 1–10 % of the argon flow each; other feed gases are not modelled.

LI-3: Stage 1 uses a volume-averaged (0D) model with an optional 1D axial profile; radial non-uniformity, 2D/3D fluid and particle-in-cell effects are not modelled.

LI-4: When oxygen is present the plasma becomes electronegative; the global model includes negative ions, but the OML probe analysis assumes an electropositive plasma, so probe results with O_2 are flagged as approximate.

LI-5: Stage 2 uses a gas-phase global model with empirical transfer of species into the liquid. It does not solve full plasma–liquid reaction–transport equations or bubble dynamics.

LI-6: Hom disinfection parameters come from published time-resolved data until validated measurements exist. An organism without published parameters cannot be simulated and is shown as unavailable.

LI-7: Measured data supplied by the laboratory may not be cited or published (supervisor decision); report validation therefore relies on published literature, and measured data are used privately for validation only.

LI-8: Agreement is targeted at ±30 % (semi-quantitative). The platform is a learning and pre-experiment planning tool, not a certified predictive instrument.

LI-9: The platform is single-user and runs on one computer; there are no user accounts, so the Student and Researcher roles are usage modes, not security boundaries.

1.6.2 Scope Evaluation Feedback and Improvements

The earlier scope (Scope Document v2.0) described a semiconductor-processing digital twin centred on machine-learning analytics. During scope evaluation, the supervisor, whose original research idea the project is, redirected it toward plasma diagnostics and environmental application, and confirmed the new parameters in written answers to two questionnaire rounds. Table 1.1 summarises the feedback and the resulting change in this SRS.

Table 1.1: Scope evaluation feedback and resulting improvements

Area	Previous scope (v2.0)	Supervisor feedback	Improvement in this SRS
Application framing	Semiconductor processing: etch rate, uniformity, process quality index, defect probability.	Remove the semiconductor framing entirely.	Platform targets plasma diagnostics and water disinfection; all semiconductor indices removed.
Plasma systems	One low-pressure Ar CCP, 1–20 mTorr.	Two plasmas: low-pressure Ar chamber and underwater high-pressure O_2 plasma.	Stage 1 (0.1–0.5 mbar, 10–100 W, 13.56 MHz) and Stage 2 (pulsed DC, 1 atm) twins.
Diagnostics	None; outputs read directly from the model.	OES on both plasmas; Langmuir probe on the low-pressure plasma.	Virtual OES (sub-module 1.2), virtual Langmuir probe (sub-module 1.3) and comparison screen M7.
Gas inputs	Gas flow and gas temperature were not physics inputs.	Ar flow 100–500 sccm; N_2 and O_2 each 5 % of the Ar flow.	Flow enters the particle balance (residence time); T_{gas} is computed; mixture modes added.
Electron energy distribution	Maxwellian rate coefficients only.	Druyvesteyn for low pressure, Maxwellian for underwater.	EEDF selectable for Stage 1 (Druyvesteyn default); Maxwellian fixed for Stage 2.

Area	Previous scope (v2.0)	Supervisor feedback	Improvement in this SRS
Spatial fidelity	0D only.	1D needed for some quantities.	1D axial profiles for Stage 1; 0D elsewhere.
Role of AI	Core of the project, aimed at semiconductor process classification (classifiers vs. baseline, anomaly detection, SHAP, recommendations).	AI optional, decided later.	Team decision (29-Oct-2026): AI kept as Core Module 2, retargeted to the new plasmas: spectrum inference, physics-anomaly detection, surrogate with SHAP and twin-verified recommendations.
Data and validation	Synthetic dataset with hidden confounders for ML; reference benchmarks.	Import measured OES/probe data for validation; data not citable; ±30 % tolerance.	Validation-only import with citation flag (sub-module 1.5); fixed ±30 % band on every comparison.

1.7 Modules

The system is a single desktop web application with **two core modules**. Core Module 1, the Digital Twin, holds all the physics: sub-modules 1.1 and 1.2 implement the supervisor's Stage 1, sub-module 1.4 implements Stage 2 and sub-module 1.3 implements Stage 3 (the Langmuir probe, which is applied to the Stage 1 plasma only, so its screens sit in the low-pressure group of the navigation). Core Module 2, the AI Module, learns from data generated by the twin and always calls the twin to verify what it proposes; the twin never depends on the AI. Two supporting components provide sessions, visualisation and export for both.

1.7.1 Core Module 1: Digital Twin (physics engine)

Sub-module 1.1 Low-Pressure Plasma Model (Stage 1)

FE-1.1.1: Configure RF power, pressure, argon flow, gas mixture, chamber geometry and model options within the supervisor envelope.

FE-1.1.2: Solve the 0D global model: particle balance for T_e and power balance for n_e.

FE-1.1.3: Compute the gas temperature from an energy balance and the species densities for Ar, N_2 and O_2 mixtures, including metastable and stepwise processes.

FE-1.1.4: Use a Druyvesteyn (default) or Maxwellian EEDF with rate coefficients integrated from published cross-sections.

FE-1.1.5: Compute 1D axial profiles between the electrodes.

FE-1.1.6: Run parameter sweeps of up to 200 points.

Sub-module 1.2 Virtual Optical Emission Spectroscopy

FE-1.2.1: Compute excited-level populations with a collisional-radiative model.

FE-1.2.2: Synthesise an emission spectrum with instrument resolution, wavelength range and noise.

FE-1.2.3: Build a Boltzmann plot and report the excitation temperature.

FE-1.2.4: Display excited-level populations against the Boltzmann distribution.

FE-1.2.5: Determine the gas temperature by N_2 rotational band or Doppler broadening.

FE-1.2.6: Estimate T_e by CR-model line ratios and analyse imported measured spectra.

Sub-module 1.3 Virtual Langmuir Probe and Diagnostics Comparison (Stage 3)

FE-1.3.1: Generate a synthetic I-V curve for a cylindrical tungsten probe with RF sheath distortion and noise.

FE-1.3.2: Analyse the curve with OML theory for V_f, V_p, T_e, n_e and λ_D, including an OML validity check.

FE-1.3.3: Extract the EEPF with the Druyvesteyn second-derivative method.

FE-1.3.4: Compare model, OES and probe values within the ±30 % tolerance.

Sub-module 1.4 Underwater Plasma Disinfection Model (Stage 2)

FE-1.4.1: Configure the pulsed DC supply, electrode gap, oxygen flow, bubble gas, water properties and target organism.

FE-1.4.2: Check pulse consistency (duty cycle and energy per pulse).

FE-1.4.3: Solve the bubble gas-phase global model with a Maxwellian EEDF.

FE-1.4.4: Transfer O_3, H_2O_2 and OH into the liquid with empirical coefficients and compute pH, ORP, conductivity and temperature change.

FE-1.4.5: Predict survival with the Hom model and report log reduction, D-value and energy yield.

FE-1.4.6: Synthesise the underwater OES and estimate n_e from H-alpha Stark broadening.

Sub-module 1.5 Validation and Measured Data (Researcher)

FE-1.5.1: Import measured OES spectra (.txt) and probe I-V curves (.csv) with column mapping and units.

FE-1.5.2: Record mandatory metadata and the citation permission for each file.

FE-1.5.3: Run literature benchmarks and measured-data comparisons with the fixed ±30 % tolerance.

FE-1.5.4: Generate a validation report.

1.7.2 Core Module 2: AI Module

Sub-module 2.1 Spectrum-to-Parameter Inference

FE-2.1.1: Infer T_e, n_e and T_{gas} from a synthetic or imported spectrum with Random Forest or XGBoost regressors.

FE-2.1.2: Give 90 % conformal prediction intervals and withhold estimates outside the training range.

FE-2.1.3: Show the physics-based analysis next to every AI estimate.

FE-2.1.4: Compare every model with a ridge-regression baseline using a paired bootstrap test.

Sub-module 2.2 Physics-Relationship Anomaly Detection

FE-2.2.1: Compute residuals between the twin's prediction and the observed diagnostics of a run or measured file.

FE-2.2.2: Score the residuals with an Isolation Forest and assign Normal, Warning or Critical severity.

FE-2.2.3: Name the residual that contributed most as the likely cause.

FE-2.2.4: Report detection quality on held-out injected faults against a range-check baseline.

Sub-module 2.3 Surrogate Model and Explainability

FE-2.3.1: Train fast surrogates of the Stage 1 and Stage 2 twins for large explorations.

FE-2.3.2: Explain outputs with global and local SHAP values.

FE-2.3.3: Check any surrogate value with the twin and show the twin value if they disagree.

Sub-module 2.4 Counterfactual Recommendation (Inverse Design)

FE-2.4.1: Accept a goal (for example, a target log reduction with the least energy) and fixed settings.

FE-2.4.2: Search the operating envelope on the surrogate and re-simulate the best candidates with the twin.

FE-2.4.3: Rank candidates by their twin-verified outcome and report what-if counterfactuals.

FE-2.4.4: State clearly when no setting inside the envelope can reach the goal.

Sub-module 2.5 AI Training, Evaluation and Model Registry

FE-2.5.1: Generate training data with the twin, with noise, faults and seed fixed before training.

FE-2.5.2: Evaluate every model on random and region-based (extrapolation) splits.

FE-2.5.3: Log every training run with MLflow and publish a model card.

FE-2.5.4: Stamp each model with the twin version and flag models trained on an older version.

1.7.3 Supporting Components

Sub-module S.1 Home, Configuration and Session Management

FE-S.1.1: Select the usage role (Student or Researcher) without login.

FE-S.1.2: Start a default experiment in one action from the quick-start list.

FE-S.1.3: Save, list, search, reopen, compare and delete sessions (configuration plus results).

FE-S.1.4: Show the load status of every reference data set and AI model.

Sub-module S.2 Visualisation, Reporting and Export

FE-S.2.1: Show every result as an interactive chart with labelled axes and SI units.

FE-S.2.2: Export results tables, figures and raw spectra or I-V data as PDF, CSV or PNG.

FE-S.2.3: Exclude non-citable measured data from every export.

FE-S.2.4: Provide a user guide and a glossary of plasma terms.

1.8 Related System Analysis/Literature Review

1.8.1 Literature Review

Global (0D) discharge models. Lieberman and Lichtenberg show that, for a low-pressure discharge, the particle balance between ionization in the volume and ion loss at the walls fixes the electron temperature independently of the absorbed power, and the power balance then fixes the electron density [1]. Lee and Lieberman extended this approach to argon and oxygen mixtures, including negative ions [2], and simulation benchmarks for capacitive discharges by Turner et al. provide reference values for testing models [8]. These models are fast enough for interactive use, which is why they are chosen here instead of fluid or particle-in-cell codes; their weakness is the lack of spatial information, which this project addresses with a 1D axial profile for Stage 1 as requested by the supervisor.

Electron energy distribution and rate data. At pressures of tens of pascals the electron energy distribution in an RF argon discharge departs from a Maxwellian, and Godyak et al. measured such distributions directly [9]. The Druyvesteyn form, which falls faster at high energy, is the supervisor's choice for low pressure, and the Druyvesteyn second-derivative method [6] is also the standard way to extract the distribution from a probe curve. Rate coefficients are obtained by integrating LXCat cross-sections [10] over the chosen distribution, the approach used by the BOLSIG+ solver [11].

Optical emission spectroscopy. Zhu and Pu developed a collisional-radiative model of the argon 1s and 2p levels valid from 1 Pa to atmospheric pressure [3] and a line-ratio method for T_e and n_e in argon–nitrogen plasmas [12]. Their work shows that 2p populations in such plasmas are not in Boltzmann equilibrium, so an excitation temperature from a Boltzmann plot must not be reported as the electron temperature; the system therefore shows both and explains the difference. Bruggeman et al. reviewed gas-temperature methods and warned that N_2 rotational temperatures in argon-rich plasmas can be distorted by energy transfer from argon metastables [13], which is why the system uses the N_2 (1,0) band at 315.9 nm with an NH overlap check and offers Doppler broadening only when the instrument resolution allows it. For dense underwater plasmas, Stark broadening of H-alpha gives n_e through the Gigosos–González–Cardeñoso relation [14].

Langmuir probes. Chen's review gives the OML theory for thin cylindrical probes and its validity limits, the effect of RF potential oscillation and practical analysis steps [5]. A

COMSATS Plasma Research Laboratory study of a low-pressure Ar–O_2 discharge used a single tungsten probe of radius 0.195 mm and 5 mm length swept from −20 to +50 V and obtained the EEDF by the Druyvesteyn method [15]; these probe dimensions and sweep settings are used as defaults because the supervisor did not specify a probe type.

Underwater plasma disinfection. Reviews by Locke et al. and Bruggeman et al. describe how discharges in and in contact with water produce O_3, H_2O_2, OH and UV, and how these species transfer into the liquid [16, 17]. A COMSATS-linked underwater capillary discharge study reported the OH (309 nm), H-alpha (656 nm) and atomic oxygen (777 and 844 nm) lines and DNA reduction of *E. coli* [18]. The Hom model [7], chosen by the supervisor, describes survival curves with shoulders or tails that a single-exponential model cannot fit. Typical municipal waste-water properties are taken from Metcalf & Eddy [19].

Machine learning on simulation data. Tree ensembles, such as Random Forests and gradient-boosted trees like XGBoost [20, 21], remain stronger than deep networks on tabular data of this kind [22], which is why the AI module uses them. Conformal prediction turns any regressor into one with prediction intervals of stated coverage [23]. Isolation Forest finds anomalies by random partitioning and needs no labelled faults [24]. SHAP attributes each prediction to its inputs with Shapley values [25], and differential evolution searches a bounded design space without gradients [26]. Because the training data come from the twin, every AI output can be checked against a known physical answer, and simple baselines (ridge regression, range checks) show whether the learning adds real value.

Gap addressed. The literature provides each model separately, but no reviewed source connects discharge settings, a solved plasma state, synthetic OES and probe signals, their analysis and a tolerance-based validation in one environment, applies the same chain to an underwater disinfection plasma, or couples it to AI that learns from the physics and is verified by it. This project fills that gap.

1.8.2 Related System Analysis

Table 1.2 compares existing systems that address parts of the problem with the proposed platform.

Table 1.2: Related system analysis with proposed project solution

Application	Weakness	Proposed Project Solution
COMSOL Multiphysics Plasma Module [27]	Expensive licence; needs finite-element expertise; no built-in synthetic OES or probe analysis workflow; slow for teaching sweeps.	Free, interactive global model with integrated virtual OES and probe; runs in seconds on a laptop.
BOLSIG+ [11]	Computes the EEDF and rate coefficients only; no plasma state, spectrum or probe curve.	Uses the same cross-section data approach inside a complete plasma-to-diagnostic chain.

Application	Weakness	Proposed Project Solution
SPECAIR [28]	Synthesises molecular spectra from user-supplied temperatures; does not compute the plasma state; argon CR kinetics not its focus.	Spectrum is computed from the solved plasma state through a CR model, so a parameter change can be followed to the spectrum.
MassiveOES [29]	Fits rotational and vibrational temperatures to measured molecular spectra; no plasma model or probe.	Combines spectrum analysis with a forward model and cross-checks with probe and model values.
Impedans Langmuir probe software [30]	Tied to vendor hardware; analyses measured curves only; cannot generate curves for teaching.	Generates RF-distorted synthetic curves and analyses both synthetic and imported curves with open methods.
PlasmaPy [31]	Python library of plasma formulas; no user interface, no integrated experiment workflow.	Point-and-click interface with guided workflow, saved sessions and reports for students.

1.9 Tools and Technologies

Table 1.3 lists the tools and technologies chosen for the project, the purpose of each, and why it was selected.

Table 1.3: Tools and technologies for the proposed project

Tool / Technology	Purpose	Rationale
Python 3.12 or later	Programming language for all models and the application	Rich scientific ecosystem; current XGBoost and SHAP releases require Python ≥ 3.12.
NumPy, SciPy	Numerical arrays; root finding (brentq), ODE integration, curve fitting, signal smoothing, differential evolution, statistical tests	Standard, well-tested numerical libraries; SciPy root finding is required for the global model.
Streamlit	Browser-based user interface served locally	Fast development of interactive scientific dashboards in pure Python.
Plotly	Interactive charts (spectra, I-V curves, profiles, survival curves)	Zoom, hover and export built in; works natively with Streamlit.
SQLite	Local storage of sessions, imported data metadata and validation results	Serverless, single-file, transactional database suited to a single-user desktop tool.
pandas	Tabular handling of imported files and results	Reliable CSV/text parsing and table export.
pytest, GitHub Actions	Unit, integration and regression tests; continuous integration	One test file per module; every requirement can be verified automatically.
Git / GitHub	Version control and collaboration	Traceable history; one commit per working sub-module.
NIST ASD, LXCat data files	Atomic line data and electron cross-sections	Authoritative, citable published data sources.
scikit-learn, XGBoost, MAPIE, SHAP	AI module: tree-ensemble regression, Isolation Forest, conformal intervals, SHAP explanations	Tree ensembles outperform deep learning on tabular data [22]; conformal intervals give coverage guarantees [23].
MLflow (local file store)	Logs every AI training run: data version, seed, hyperparameters, metrics	Reproducible, queryable training history with no server [32].

Tool / Technology	Purpose	Rationale
HTML, CSS, JavaScript	Interactive mockup prototype used for requirement analysis	Clickable prototype opens in any browser with no installation.

1.10 Project Contribution

The project contributes the following, compared with existing manual practice and separate tools:

- **One plasma state for every diagnostic.** The spectrum, the probe curve and the comparison table are all computed from the same solved plasma state, so the user can trace any feature in a diagnostic signal back to its physical cause.
- **Physics instead of curve fitting.** Electron temperature and density come from particle and power balance equations, and every other output is derived from them; no output is set independently or fitted to look right.
- **Built-in cross-validation.** OES, probe and model values are compared automatically against a tolerance fixed before the comparison, which teaches that two diagnostics measuring the same quantity must agree.
- **Honest treatment of non-equilibrium.** The system shows the excitation temperature and the CR-model populations together and explains why the excitation temperature is not the electron temperature.
- **Two plasmas in one tool.** The same diagnostic framework covers a low-pressure argon plasma and an underwater disinfection plasma, linking discharge settings to bacterial log reduction and energy yield.
- **Responsible use of measured data.** Measured files are used for validation only, never for tuning, and non-citable data are removed from every export automatically.
- **AI that learns from physics and is checked by it.** The AI module is trained only on twin-generated data, every AI estimate is shown beside the physics result, and every recommendation is re-simulated by the twin before it is displayed.
- **Honest AI evaluation.** Every model is compared with a simple baseline using a statistical test, evaluated on a held-out region of the envelope as well as a random split, and dataset settings are fixed before training.
- **Anomalies defined by physics.** The anomaly detector looks for diagnostics that disagree with the twin, so it catches faults that a range check cannot see.
- **Local-laboratory defaults.** Probe dimensions, sweep settings and chamber geometry follow the COMSATS Plasma Research Laboratory, so a student can rehearse the real experiment before entering the laboratory.

1.11 Relevance to Course Modules

- **Software Engineering and Software Requirements Engineering:** requirement elicitation by supervisor questionnaires, mockup-based requirement analysis, event–response modelling and traceability.
- **Human-Computer Interaction:** user classes, interactive prototype, usability and accessibility requirements.
- **Database Systems:** session and validation data model in SQLite with transactions and parameterised queries.
- **Numerical Computing, Calculus and Differential Equations:** root finding for the particle balance, ODE integration for species kinetics, numerical differentiation of probe curves.
- **Probability and Statistics:** least-squares fits with uncertainty, noise models, tolerance-based validation and prediction intervals.
- **Applied Physics:** plasma physics, atomic spectroscopy and electrostatics behind every model.
- **Artificial Intelligence and Machine Learning (Core Module 2):** tree-ensemble regression, unsupervised anomaly detection, conformal uncertainty, SHAP explainability, surrogate modelling and optimisation-based inverse design.
- **Software Quality Engineering:** pytest unit and regression tests, continuous integration and measurable NFRs.

Chapter 2: Requirement Analysis

This chapter specifies what the system must do and how well it must do it. It describes the user classes, the operating environment and the design constraints, presents the use case diagram, explains the mockup-based and event–response techniques used to identify requirements, lists the functional requirements screen by screen together with their business rules and backend events, and finally states measurable non-functional and external interface requirements. All values shown in mockups are illustrative placeholders, not computed results.

2.1 User Classes and Characteristics

Table 2.1 describes the user classes (human actors) and the external actors that interact with the system. There is no login: the two human classes are usage roles selected on the Home screen.

Table 2.1: User classes and external actors

User class / Actor	Description and characteristics
Student (Learner)	An undergraduate or graduate physics student learning plasma diagnostics, the primary and most frequent user. Has basic plasma physics knowledge but limited laboratory time and little programming experience. Uses Stage 1 and Stage 2 simulations, virtual OES and probe, comparison, AI inference, anomaly checks, the surrogate and recommendations, sessions and export. Needs guided defaults, clear units and explanations of every symbol.
Researcher	A faculty member, laboratory engineer or research student (including the supervisor) who has, or will have, measured data. Inherits every Student capability and additionally imports measured OES and probe files, manages citation permission, runs validation against measured data and literature benchmarks, and trains and evaluates the AI models. Expects transparent methods and exportable validation reports.
OES Spectrometer (external)	The laboratory high-resolution spectrometer (better than 0.01 nm). It does not connect to the system; its vendor software exports spectra as text files that a Researcher imports.
Langmuir Probe System (external)	The laboratory probe and its acquisition software. It exports I-V curves as .csv files that a Researcher imports.
Reference Data Source (external)	Published data sets shipped with the application: NIST atomic line data, LXCat cross-sections, N_2 band constants and published benchmark and disinfection data. Read at start-up and during validation.

2.2 Operating Environment

OE-1: The system shall run on a 64-bit desktop or laptop computer with Windows 10/11 or Ubuntu 22.04 LTS or later, at least 4 GB RAM (8 GB recommended), 2 GB free disk space and a display of at least 1366 × 768 pixels.

OE-2: The system shall run on Python 3.12 or later with the exact package versions pinned in requirements.txt.

OE-3: The user interface shall be served locally and shall work in the two most recent major versions of Google Chrome, Microsoft Edge and Mozilla Firefox.

OE-4: The system shall operate without an internet connection; all reference data are installed with the application.

OE-5: Measured instrument data shall reach the system only as files (.txt or .csv) copied from the instrument computers by USB drive or network share.

OE-6: Each installation shall serve one user at a time and shall store its data in a local SQLite file inside the application data folder.

2.3 Design and Implementation Constraints

CON-1: The system shall be implemented in Python 3.12+ with Streamlit, Plotly, SQLite, NumPy and SciPy, and, for the AI module, scikit-learn, XGBoost, MAPIE, SHAP and MLflow (local tracking only); no substitute technology or new dependency may be introduced without supervisor approval.

CON-2: The Stage 1 plasma model shall be a volume-averaged global model solved by particle and power balance, using SciPy root finding for T_e and then solving for n_e; every derived output shall be computed from T_e and n_e and never set independently.

CON-3: Operating envelopes are fixed by the supervisor: Stage 1 — 10–100 W, 0.1–0.5 mbar, 13.56 MHz CCP, Ar 100–500 sccm, N_2 and O_2 at 5 % of the Ar flow by default; Stage 2 — pulsed DC up to 10 kV, 5–10 kHz, 100 W average, pin-to-pin gap 1–3 mm, 1 atm, O_2 100–300 sccm.

CON-4: The Stage 1 EEDF shall default to Druyvesteyn (Maxwellian selectable); the Stage 2 EEDF shall be Maxwellian.

CON-5: Low-pressure n_e for OES analysis shall be taken from the Langmuir probe value (simulated, or measured when imported); underwater n_e shall be obtained from H-alpha Stark broadening.

CON-6: Measured data shall be used for validation only and shall never be used to fit or tune model parameters; measured data are non-citable unless the Researcher records permission.

CON-7: The agreement tolerance shall be ±30 % and shall be fixed in a versioned configuration file before any comparison is run.

CON-8: One-dimensional modelling shall be used only for Stage 1 spatial profiles; all other models are 0D.

CON-9: Atomic data, cross-sections, rate coefficients and benchmark values shall come from published sources whose citation is stored with the data.

CON-10: The AI module is a core module. Its models shall be tree ensembles (Random Forest, XGBoost) or Isolation Forest, with no deep learning, trained only on synthetic data generated by the digital twin, with dataset noise, fault settings and seed fixed before training. Measured data may be used to test AI outputs but never to train them.

CON-11: The system shall not drive or communicate with laboratory hardware.

CON-12: Every AI recommendation shall be re-simulated through the digital twin before it is shown, and every AI or surrogate value shall be labelled as such and displayed next to the physics result.

2.4 Use Case Diagram

Figure 2.1 shows the use case diagram. The Student is the primary actor; the Researcher specialises the Student and adds measured-data import and validation. The OES Spectrometer and Langmuir Probe System are external actors that supply files, and the Reference Data Source supplies published data. «include» marks a step that always happens inside the base use case (for example, validation of the configuration before every run, or re-simulating a recommendation with the digital twin), and «extend» marks optional behaviour (a parameter sweep). The AI module's use cases form their own group; training and evaluating AI models is a Researcher use case. Table 2.2 summarises every use case and traces it to its mockup and functional requirements.

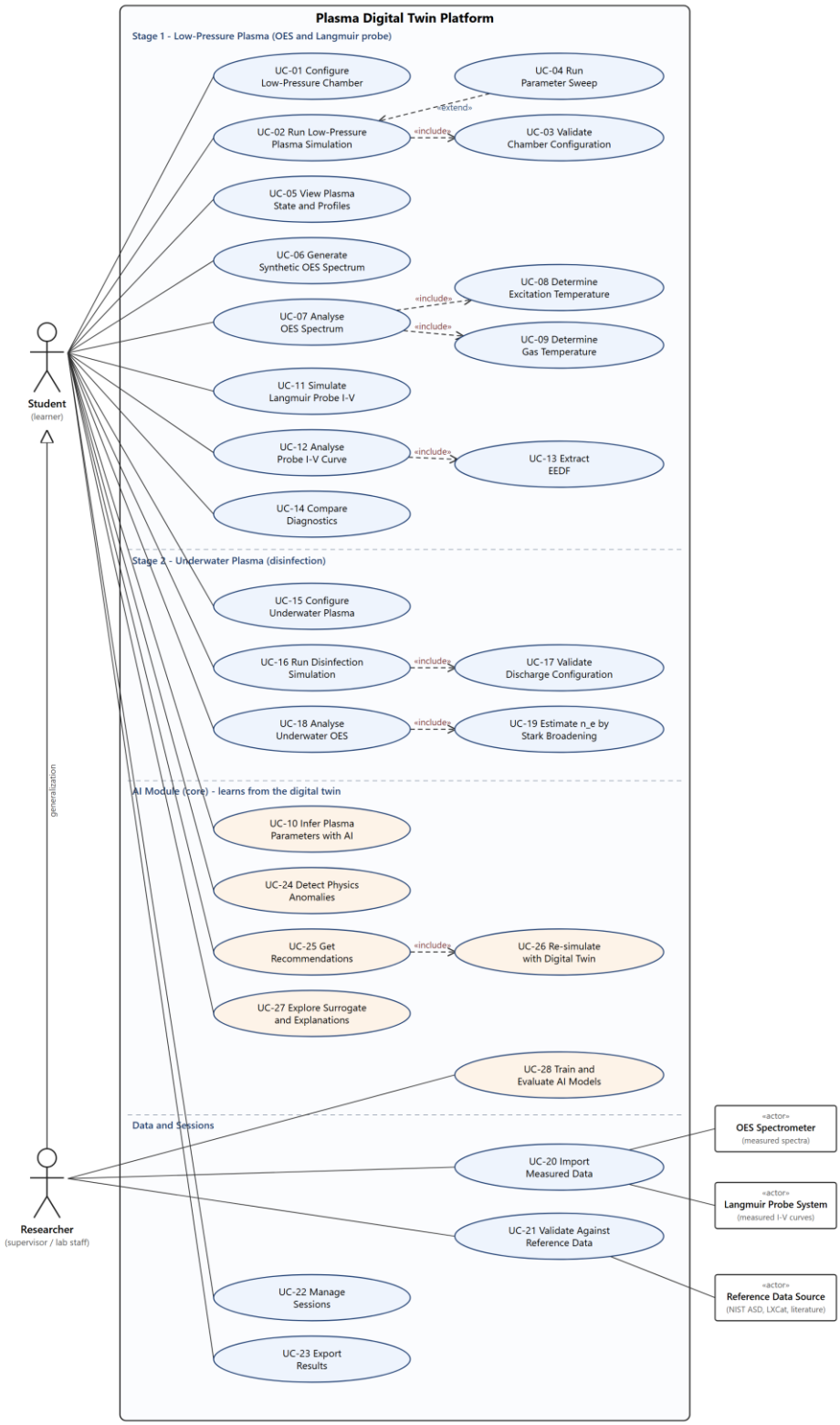


Figure 2.1: Use case diagram of the Plasma Digital Twin Platform

Table 2.2: Use case summary and traceability

UC-ID	Use case	Primary actor	Goal	Mockup	FRs
UC-01	Configure Low-Pressure Chamber	Student	Set power, pressure, flow, mixture, geometry and model options.	M2	FR2.1–FR2.10, FR2.13
UC-02	Run Low-Pressure Plasma Simulation	Student	Solve the Stage 1 model for the current configuration.	M2, M3	FR2.11, FR17.2
UC-03	Validate Chamber Configuration (include)	—	Check every input against the envelope before a run.	M2	FR2.10, FR17.3
UC-04	Run Parameter Sweep (extend UC-02)	Student	Repeat the simulation over a range of one parameter.	M2, M3	FR2.12, FR3.7, FR17.14
UC-05	View Plasma State and Profiles	Student	Read T_e, n_e, T_{gas}, species, EEDF, profile and balances.	M3	FR3.1–FR3.6
UC-06	Generate Synthetic OES Spectrum	Student	Compute the spectrum with instrument settings.	M4	FR4.1–FR4.5, FR4.7
UC-07	Analyse OES Spectrum	Student	Analyse a synthetic or imported spectrum.	M5	FR5.1, FR5.5, FR5.7–FR5.10
UC-08	Determine Excitation Temperature (include)	—	Boltzmann plot and T_{exc}.	M5	FR5.2–FR5.4
UC-09	Determine Gas Temperature (include)	—	N_2 rotational or Doppler T_{gas}.	M5	FR5.6, FR17.8
UC-11	Simulate Langmuir Probe I-V	Student	Generate an RF-distorted I-V curve.	M6	FR6.1–FR6.5
UC-12	Analyse Probe I-V Curve	Student	OML analysis for V_f, V_p, T_e, n_e, λ_D.	M6	FR6.6, FR6.7, FR6.10
UC-13	Extract EEDF (include)	—	Druyvesteyn second-derivative EEPF.	M6	FR6.8
UC-14	Compare Diagnostics	Student	Model vs OES vs probe within ±30 %.	M7	FR7.1–FR7.6
UC-15	Configure Underwater Plasma	Student	Set pulse, gap, gas, water and organism.	M8	FR8.1–FR8.12
UC-16	Run Disinfection Simulation	Student	Solve Stage 2 and predict survival.	M8, M9	FR8.13, FR9.1–FR9.3, FR9.5
UC-17	Validate Discharge Configuration (include)	—	Duty cycle and envelope checks.	M8	FR8.4, FR17.5
UC-18	Analyse Underwater OES	Student	Underwater spectrum with OH, H-alpha, O I lines.	M9	FR9.4
UC-19	Estimate n_e by Stark Broadening (include)	—	n_e from H-alpha Stark width.	M9	FR9.4
UC-20	Import Measured Data	Researcher; OES Spectrometer; Langmuir Probe System	Upload, map and describe measured files.	M10, M4, M6	FR10.1–FR10.5, FR4.6, FR6.9
UC-21	Validate Against Reference Data	Researcher; Reference Data Source	Benchmarks and measured-data comparisons.	M10	FR10.6–FR10.8
UC-22	Manage Sessions	Student	Save, list, reopen, compare, delete sessions.	M11, M1	FR11.1–FR11.4, FR1.3, FR3.9
UC-23	Export Results	Student	Export tables, figures and data.	M11	FR11.5, FR3.10, FR4.7, FR9.6
UC-10	Infer Plasma Parameters with AI	Student	AI estimate of T_e, n_e, T_{gas} with intervals.	M12	FR12.1–FR12.6, FR17.18
UC-24	Detect Physics Anomalies	Student	Flag runs whose diagnostics break the physics.	M13	FR13.1–FR13.7, FR17.20

UC-ID	Use case	Primary actor	Goal	Mockup	FRs
UC-25	Get Recommendations	Student	Settings for a goal, ranked by twin outcome.	M14	FR14.1–FR14.8, FR17.21
UC-26	Re-simulate with Digital Twin (include)	—	Verify each candidate with the twin.	M14	FR14.3, FR14.6
UC-27	Explore Surrogate and Explanations	Student	Fast maps and SHAP explanations.	M15	FR15.1–FR15.6, FR17.22
UC-28	Train and Evaluate AI Models	Researcher	Train, evaluate and register models.	M16	FR16.1–FR16.5, FR17.19

2.5 Requirement Identifying Technique

Two complementary techniques were used. The first is **mockup-based requirement analysis**. Acting as UI/UX designers for a separate implementation team, the authors broke each feature into screens, gave every screen a mockup identifier (M1 to M16), and built them as a clickable HTML prototype (Appendix A) so the supervisor could walk through the workflow. Each interactive element on each screen (input, toggle, button, chart, table or message) was then turned into a feature-level functional requirement, and the policy that governs its behaviour was recorded as a business rule. Domain values in the business rules come from the supervisor's written questionnaire answers; where the supervisor left an item blank, a published source is cited (Appendix C lists every value and its source).

The second technique is the **event–response table**. Much of this system's behaviour happens behind the screens: solving the plasma models, checking convergence, loading reference data, parsing files, applying the citation filter and enforcing the tolerance. These backend behaviours were identified as events, and for each event the trigger, source, system state before, response, system state after and exception handling were recorded (Section 2.6.17). Functional requirements were then derived from the events (Section 2.6.18).

Requirement identifiers follow the pattern FRm.n, where m is the mockup number (1–16) and n the requirement on that screen; m = 17 is used for requirements derived from backend events. Every requirement is written so that it can be tested, and the traceability matrix in Appendix B links business objectives, modules, use cases, mockups and requirements.

2.6 Functional Requirements

The functional requirements are listed below by mockup. Each subsection shows the mockup screen as a figure, followed by the requirements derived from it in the department's recommended format (Feature | FR-ID: Statement | Business Rule).

2.6.1 Mockup M1 – Home Screen

The Home screen is the entry point. It offers the four work areas (Stage 1, Stage 2, Diagnostics & Data and the AI Module), a one-step quick start, the most recent sessions, the usage role, the status of the reference data and AI models, and help. Figure 2.2 shows the mockup and Table 2.3 lists the functional requirements derived from it.

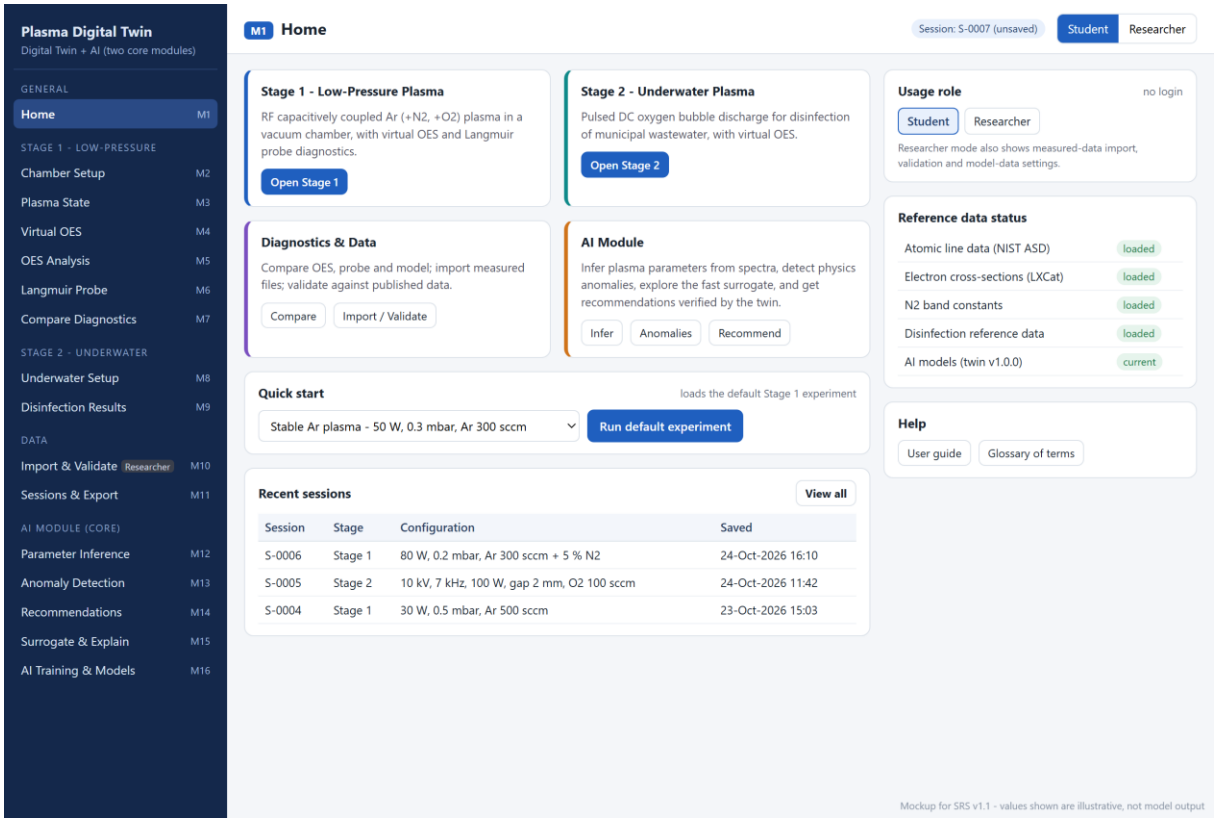


Figure 2.2: Mockup M1 – Home Screen

Table 2.3: Functional requirements derived from Mockup M1 – Home Screen

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Stage navigation	FR1.1: The system shall display entry points to Stage 1 (Low-Pressure Plasma), Stage 2 (Underwater Plasma), Diagnostics & Data and the AI Module, each opening its first screen.	Stage 1 is listed first because the supervisor ranks it as the most important stage.
Quick start	FR1.2: The system shall allow the user to choose a predefined example experiment and run it in a single action.	Every quick-start configuration shall lie inside the operating envelope; the default is 50 W, 0.3 mbar, Ar 300 sccm.
Recent sessions	FR1.3: The system shall list the most recently saved sessions with session ID, stage, configuration summary and save time.	At most three sessions, newest first; "View all" opens M11.
Usage role	FR1.4: The system shall allow the user to switch between the Student and Researcher roles without logging in.	Researcher mode additionally shows measured-data import, overlays and validation; the role is a usage mode, not a security control (LI-9).
Reference data status	FR1.5: The system shall display the load status of each reference data set (NIST atomic lines, LXCat cross-sections, N_2 band constants, disinfection reference data) and of the AI models.	A data set marked missing or corrupt disables every feature that depends on it (event E6); an out-of-date AI model is shown as such (event E22).

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Help	FR1.6: The system shall provide a user guide and a glossary of the plasma terms and symbols used on the screens.	The glossary shall define every symbol displayed anywhere in the application.

2.6.2 Mockup M2 – Low-Pressure Chamber Configuration

This screen configures the Stage 1 discharge, the chamber geometry and the model options, and optionally a parameter sweep. Figure 2.3 shows the mockup and Table 2.4 lists the functional requirements derived from it.

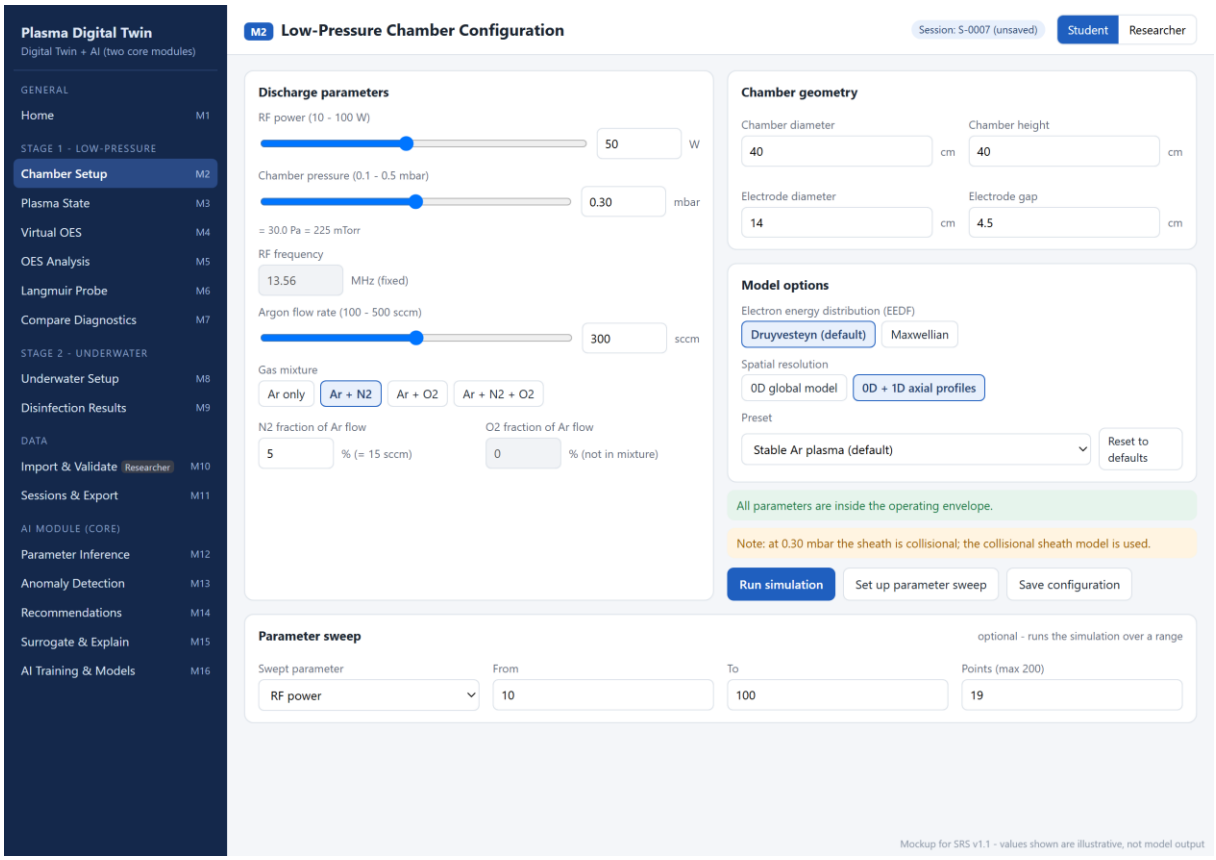


Figure 2.3: Mockup M2 – Low-Pressure Chamber Configuration

Table 2.4: Functional requirements derived from Mockup M2 – Low-Pressure Chamber Configuration

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
RF power	FR2.1: The system shall accept the RF power through a linked slider and numeric field.	Range 10–100 W, step 1 W (supervisor envelope).
Chamber pressure	FR2.2: The system shall accept the chamber pressure in mbar and display the equivalent value in Pa and mTorr.	Range 0.1–0.5 mbar; 1 mbar = 100 Pa = 750.06 mTorr.
RF frequency	FR2.3: The system shall display the RF frequency as a fixed, non-editable value.	13.56 MHz for the CCP (supervisor).

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Argon flow rate	FR2.4: The system shall accept the argon flow rate through a linked slider and numeric field.	Range 100–500 sccm (supervisor).
Gas mixture	FR2.5: The system shall allow the user to choose the mixture mode (Ar only, Ar + N_2, Ar + O_2, Ar + N_2 + O_2) and to enter each admixture as a percentage of the argon flow, showing the resulting flow in sccm.	Default 5 % of the Ar flow for each admixture (supervisor); allowed 1–10 %; fields of gases not in the mixture are disabled and set to 0.
Chamber geometry	FR2.6: The system shall accept the chamber diameter and height, electrode diameter and electrode gap.	Defaults 40, 40, 14 and 4.5 cm; electrode diameter < chamber diameter; gap < chamber height; all values > 0.
EEDF	FR2.7: The system shall allow the user to select a Druyvesteyn or Maxwellian electron energy distribution.	Druyvesteyn is the default for Stage 1 (supervisor).
Spatial resolution	FR2.8: The system shall allow the user to choose the 0D global model or the 0D model with 1D axial profiles.	1D is offered only for Stage 1 (CON-8).
Preset and reset	FR2.9: The system shall load a named preset and shall reset all fields to their defaults on request.	Presets are fixed, documented configurations inside the envelope.
Input validation	FR2.10: The system shall validate every parameter when it changes and display the envelope status and any model-regime note (for example, collisional sheath).	Run is disabled while any input is invalid; messages state the allowed range (event E3).
Run simulation	FR2.11: The system shall run the Stage 1 simulation for the current configuration and open the results screen (M3).	Runs only after FR2.10 passes.
Parameter sweep	FR2.12: The system shall allow the user to define a sweep by choosing the swept parameter (RF power, pressure, argon flow or admixture fraction), the start and end values and the number of points.	Start < end, both inside the envelope; at most 200 points; all other parameters held constant.
Save configuration	FR2.13: The system shall save the current configuration under a user-given name for later reuse.	Names must be unique; saving does not run the model.

2.6.3 Mockup M3 – Low-Pressure Plasma State Results

This screen presents the solved Stage 1 plasma state, the distribution and profile used, the energy balance and any sweep results. Figure 2.4 shows the mockup and Table 2.5 lists the functional requirements derived from it.

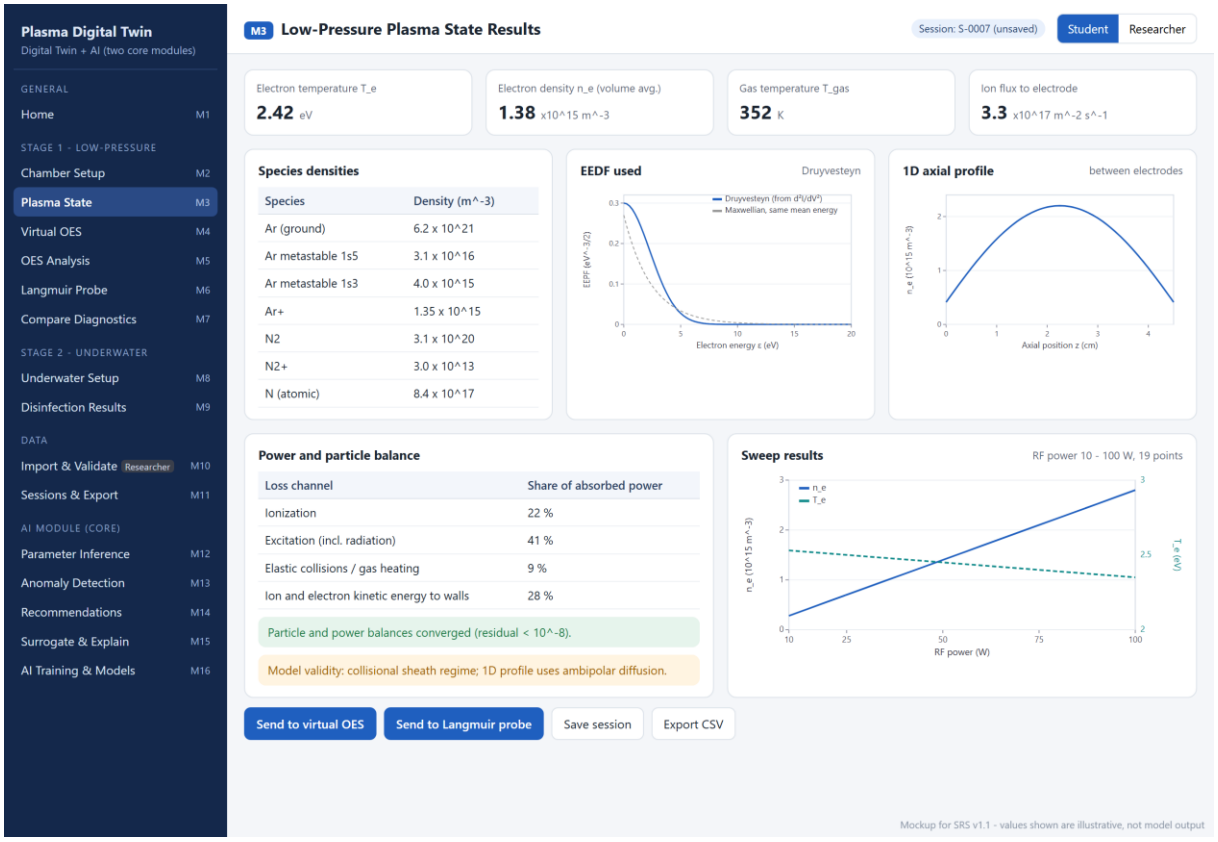


Figure 2.4: Mockup M3 – Low-Pressure Plasma State Results

Table 2.5: Functional requirements derived from Mockup M3 – Low-Pressure Plasma State Results

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Key results	FR3.1: The system shall display T_e, volume-averaged n_e, T_{gas} and the ion flux to the electrode with units.	All values derive from the solved T_e and n_e (CON-2); none is set independently.
Species densities	FR3.2: The system shall display the densities of ground-state Ar, Ar metastables ($1s_5$, $1s_3$), Ar^+ and the admixture species present in the selected mixture.	Species of gases not in the mixture are not shown.
EEDF plot	FR3.3: The system shall plot the electron energy distribution used in the run.	Shows the selected form (FR2.7).
1D profile	FR3.4: The system shall plot the axial n_e profile between the electrodes when 1D profiles were selected.	Hidden for 0D-only runs.
Power and particle balance	FR3.5: The system shall display the share of absorbed power lost through each channel and the convergence status of the balances.	Shares sum to 100 ± 0.5 %; a run is accepted only if the residual is below 10^{-8} (ACC-3).
Model validity	FR3.6: The system shall display the regime assumptions used in the run (sheath model, diffusion model).	Assumptions are generated from the configuration, not typed by hand.
Sweep results	FR3.7: The system shall plot n_e and T_e against the swept parameter after a sweep.	Dual axes, each labelled with symbol and unit.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Send to diagnostics	FR3.8: The system shall pass the current plasma state to the virtual OES (M4) and the virtual Langmuir probe (M6).	Diagnostics always use the state of the open session, whose ID is displayed.
Save session	FR3.9: The system shall save the configuration and results as a session with a unique ID.	IDs are sequential (S-0001, S-0002, …) and never reused.
Export CSV	FR3.10: The system shall export the results of the run as a CSV file.	The file header records the session ID, configuration and software version.

2.6.4 Mockup M4 – Virtual OES – Synthetic Spectrum

This screen synthesises the emission spectrum of the current Stage 1 plasma and lists the identified lines. Figure 2.5 shows the mockup and Table 2.6 lists the functional requirements derived from it.

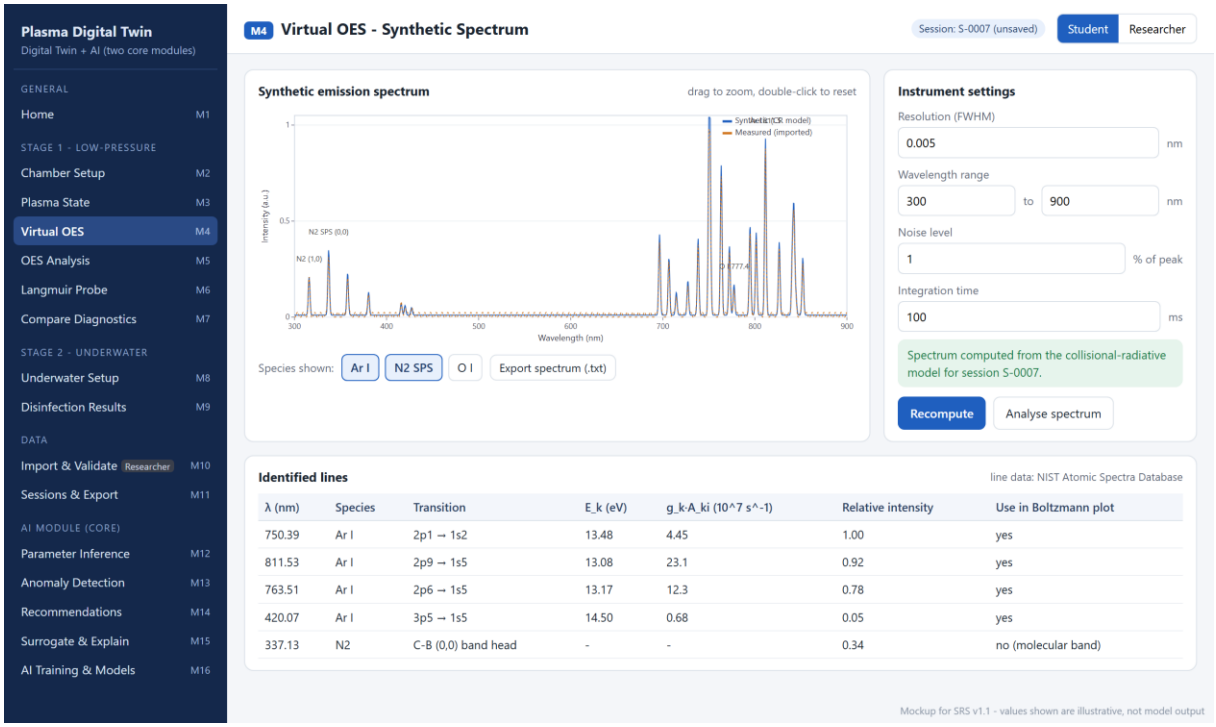


Figure 2.5: Mockup M4 – Virtual OES – Synthetic Spectrum

Table 2.6: Functional requirements derived from Mockup M4 – Virtual OES – Synthetic Spectrum

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Synthetic spectrum	FR4.1: The system shall compute and display the emission spectrum of the current plasma state from collisional-radiative level populations, with interactive zoom.	Intensity of each line is proportional to the upper-level population × A_{ki} / λ; line data from NIST ASD.
Species toggles	FR4.2: The system shall allow the user to show or hide the Ar I, N_2 second positive system and O I contributions.	Only species present in the mixture can be enabled.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Instrument settings	FR4.3: The system shall accept the instrument resolution (FWHM), wavelength range, noise level and integration time.	Resolution 0.001–2 nm (default 0.005 nm; the laboratory instrument is better than 0.01 nm); range within 200–1000 nm; noise 0–20 % of peak.
Recompute	FR4.4: The system shall recompute the spectrum when the user changes the instrument settings and requests it.	The plasma state is not re-solved; only the instrument model is reapplied.
Identified lines	FR4.5: The system shall list the identified lines with wavelength, species, transition, upper-level energy, g_kA_{ki}, relative intensity and whether each line is used in the Boltzmann plot.	Molecular bands are excluded from the Boltzmann plot.
Overlay measured	FR4.6: The system shall allow a Researcher to overlay an imported measured spectrum on the synthetic spectrum.	Only Stage 1 spectra imported through M10; both spectra normalised to the same reference line.
Export spectrum	FR4.7: The system shall export the spectrum as a two-column text file (wavelength, intensity).	Header records session ID and instrument settings.
Analyse spectrum	FR4.8: The system shall open the OES analysis screen (M5) with the current spectrum.	—

2.6.5 Mockup M5 – OES Analysis – Boltzmann Plot and Temperatures

This screen analyses a synthetic or imported spectrum to obtain the excitation temperature, the level populations, the gas temperature and the electron density used. Figure 2.6 shows the mockup and Table 2.7 lists the functional requirements derived from it.

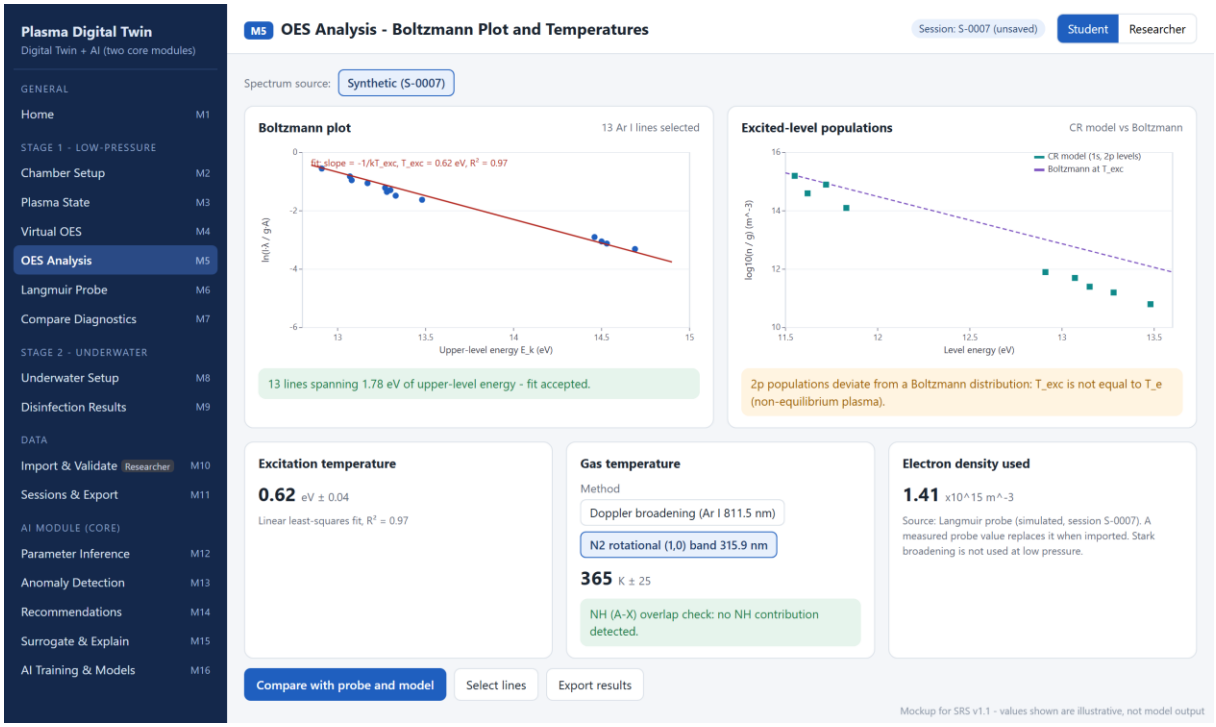


Figure 2.6: Mockup M5 – OES Analysis – Boltzmann Plot and Temperatures

Table 2.7: Functional requirements derived from Mockup M5 – OES Analysis – Boltzmann Plot and Temperatures

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Spectrum source	FR5.1: The system shall allow the user to analyse either the synthetic spectrum of the open session or (Researcher) an imported measurement.	Imported spectra must have complete metadata (FR10.2).
Boltzmann plot	FR5.2: The system shall plot $\ln(I\lambda/g_kA_{ki})$ against upper-level energy for the selected Ar I lines and fit a straight line.	At least 4 lines spanning at least 1.0 eV are required; otherwise the fit is not reported (event E9).
Excitation temperature	FR5.3: The system shall report $T_{exc} = -1/\text{slope}$ with its standard error and the R^2 of the fit.	A fit with $R^2 < 0.90$ is flagged as unreliable.
Level populations	FR5.4: The system shall display the CR-model populations of the excited levels against a Boltzmann distribution and state when the plasma is not in equilibrium.	T_{exc} is never presented as T_e.
T_e by line ratio	FR5.5: The system shall estimate T_e by matching measured line-intensity ratios to CR-model predictions.	Result is displayed in the comparison screen (M7).
Gas temperature	FR5.6: The system shall determine T_{gas} by the method selected: N_2 rotational temperature from the (1,0) band at 315.9 nm, or Doppler broadening of the Ar I 811.5 nm line.	N_2 method only when N_2 is in the mixture, with an NH (A–X) overlap check; Doppler only when the instrument FWHM ≤ expected Doppler FWHM (event E8).
Electron density used	FR5.7: The system shall display the n_e used in the OES analysis and its source.	Source is the Langmuir probe value — simulated, or measured when imported (supervisor); Stark broadening is not used at low pressure.
Select lines	FR5.8: The system shall allow the user to include or exclude individual lines and shall refit immediately.	Excluded lines are listed in the export.
Compare	FR5.9: The system shall open the comparison screen (M7) with the analysis results.	—
Export results	FR5.10: The system shall export the analysis results and the fitted line list.	CSV with method and uncertainty columns.

2.6.6 Mockup M6 – Virtual Langmuir Probe

This screen generates the I-V curve of a cylindrical probe in the current Stage 1 plasma and analyses it. Figure 2.7 shows the mockup and Table 2.8 lists the functional requirements derived from it.

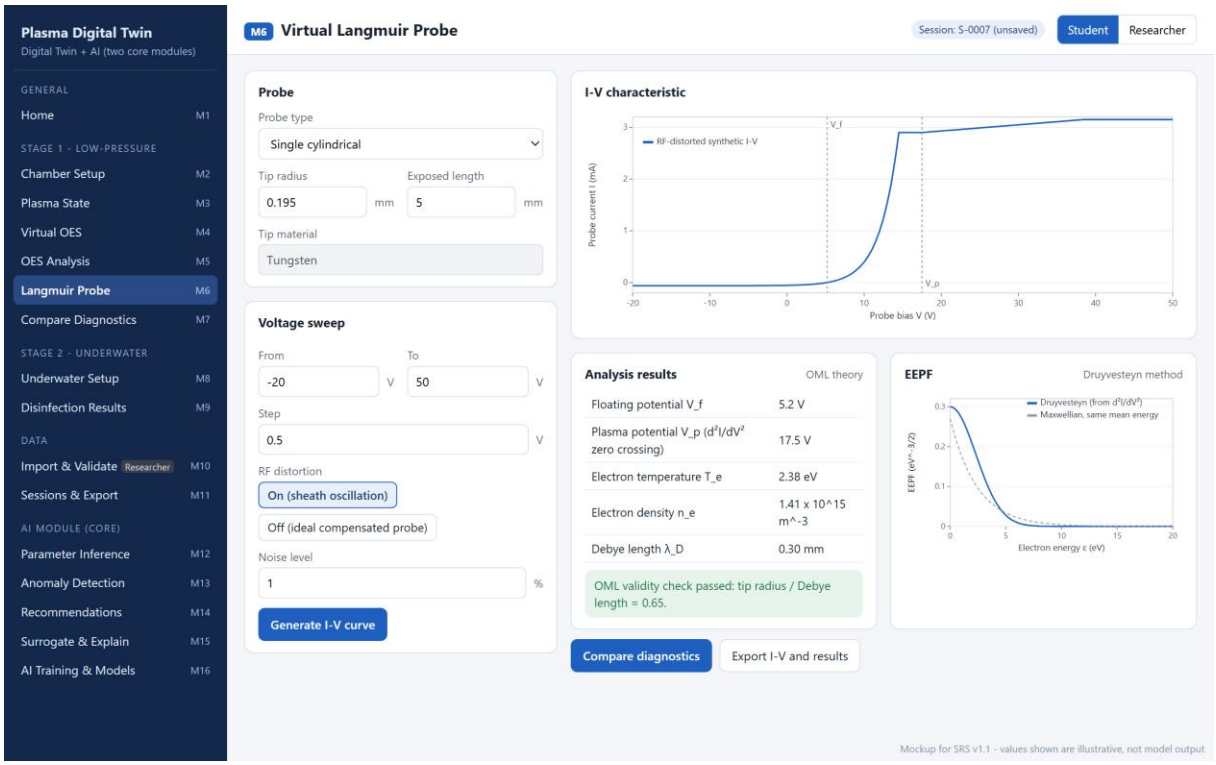


Figure 2.7: Mockup M6 – Virtual Langmuir Probe

Table 2.8: Functional requirements derived from Mockup M6 – Virtual Langmuir Probe

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Probe geometry	FR6.1: The system shall accept the probe type, tip radius and exposed length; the tip material is shown as tungsten.	Single cylindrical probe; defaults 0.195 mm and 5 mm (COMSATS laboratory probe); radius 0.05–1 mm; length 1–20 mm.
Voltage sweep	FR6.2: The system shall accept the sweep start, end and step voltages.	Defaults −20 V to +50 V in 0.5 V steps; limits −100 to +100 V; step 0.1–2 V; at most 2001 points.
RF distortion	FR6.3: The system shall allow RF sheath-oscillation distortion of the I-V curve to be switched on or off.	On = time average over the 13.56 MHz oscillation of the plasma potential (supervisor); Off = ideal compensated probe.
Noise level	FR6.4: The system shall accept a noise level for the synthetic current.	0–10 % of the electron saturation current; seeded for reproducibility.
Generate I-V curve	FR6.5: The system shall generate the synthetic I-V curve from the current plasma state and display it.	Uses T_e, n_e and the EEDF of the open session.
Analysis results	FR6.6: The system shall determine V_f (zero current), V_p (zero crossing of d^2I/dV^2), T_e (slope of ln I_e in the retardation region), n_e (OML ion current) and λ_D.	OML theory (supervisor); results flagged as approximate when O_2 is present (LI-4).

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
OML validity check	FR6.7: The system shall compute r_p/λ_D and report whether the OML assumption holds.	OML accepted when $r_p/\lambda_D \leq 1$; otherwise a warning is shown (event E12).
EEPF	FR6.8: The system shall extract the EEPF from the second derivative of the I-V curve and plot it with a Maxwellian of the same mean energy.	Druyvesteyn method; the smoothing window is shown with the result.
Import measured I-V	FR6.9: The system shall allow a Researcher to analyse an imported measured I-V curve with the same method.	Only .csv files imported through M10.
Compare / export	FR6.10: The system shall open the comparison screen (M7) and export the I-V curve and results.	CSV with V, I and analysis summary.

2.6.7 Mockup M7 – Diagnostics Comparison (Model vs OES vs Probe)

This screen places the model, OES and probe values side by side and judges their agreement. Figure 2.8 shows the mockup and Table 2.9 lists the functional requirements derived from it.

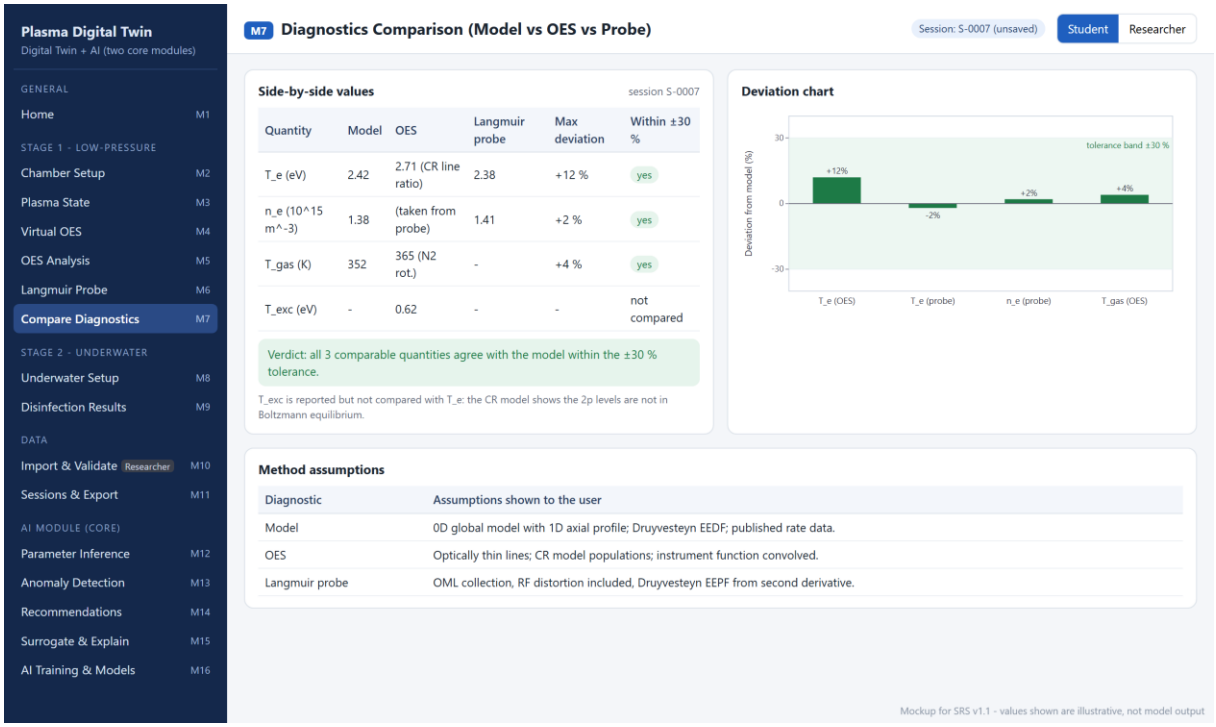


Figure 2.8: Mockup M7 – Diagnostics Comparison (Model vs OES vs Probe)

Table 2.9: Functional requirements derived from Mockup M7 – Diagnostics Comparison (Model vs OES vs Probe)

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Side-by-side values	FR7.1: The system shall display T_e, n_e and T_{gas} from the model, OES and probe with the maximum deviation and a pass/fail flag.	Deviation = (diagnostic − model)/model; pass when |deviation| ≤ 30 % (supervisor); the tolerance is not editable on screen.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Dependent values	FR7.2: The system shall mark a value that one diagnostic takes from another (n_e in OES taken from the probe) and exclude it from the independent comparison.	Prevents a value being counted as agreeing with itself.
Verdict	FR7.3: The system shall show an overall verdict stating how many comparable quantities agree within tolerance.	Text label as well as colour (USE-5).
Deviation chart	FR7.4: The system shall plot the deviation of each diagnostic value with the ±30 % band.	—
Excitation temperature	FR7.5: The system shall display T_{exc} but exclude it from the comparison with T_e and explain why.	Non-equilibrium 2p populations (FR5.4).
Method assumptions	FR7.6: The system shall list the assumptions of each method (model, OES, probe).	Generated from the settings actually used.

2.6.8 Mockup M8 – Underwater Plasma Configuration

This screen configures the Stage 2 pulsed DC bubble discharge, the waste water and the target organism. Figure 2.9 shows the mockup and Table 2.10 lists the functional requirements derived from it.

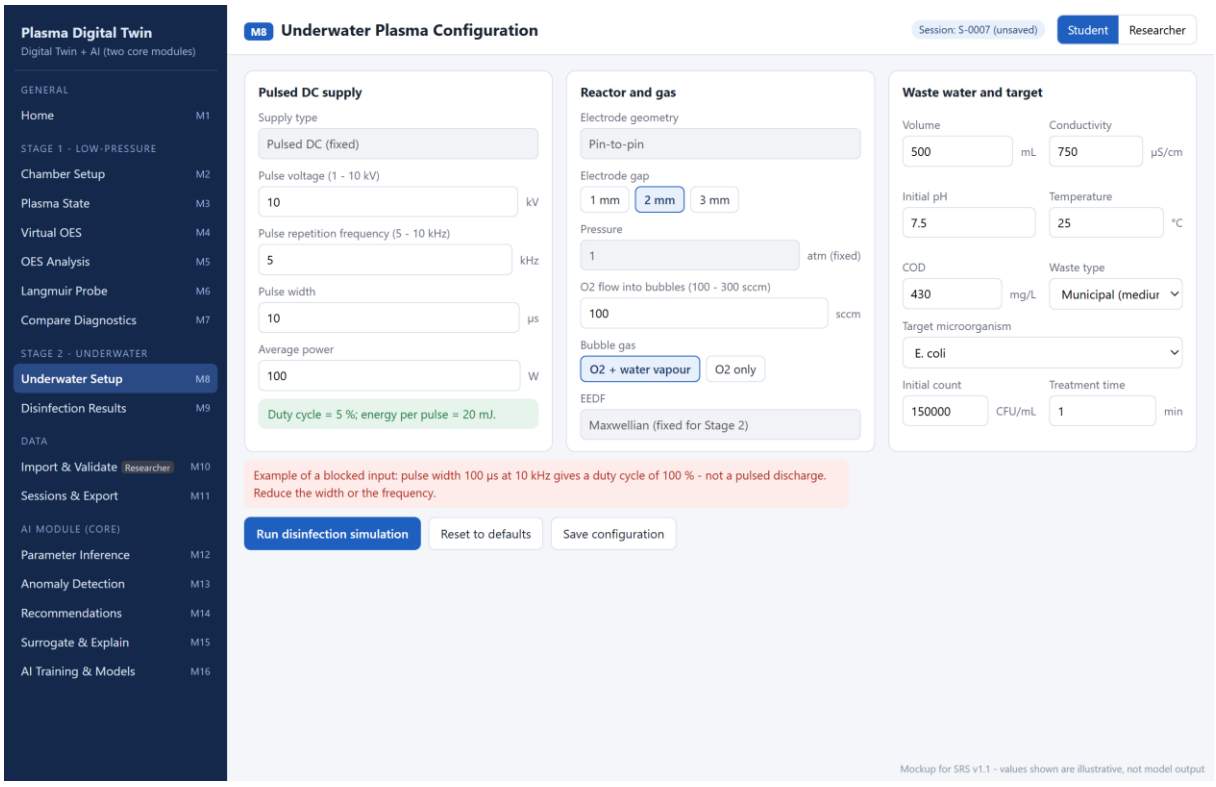


Figure 2.9: Mockup M8 – Underwater Plasma Configuration

Table 2.10: Functional requirements derived from Mockup M8 – Underwater Plasma Configuration

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Supply type	FR8.1: The system shall display the supply type as pulsed DC (fixed).	Supervisor.
Pulse voltage	FR8.2: The system shall accept the pulse voltage.	Range 1–10 kV; default 10 kV (supervisor).
Pulse repetition frequency	FR8.3: The system shall accept the pulse repetition frequency.	Range 5–10 kHz (supervisor).
Pulse width and duty cycle	FR8.4: The system shall accept the pulse width in µs and shall compute and display the duty cycle.	Duty cycle = pulse width × frequency must be < 1, otherwise the run is blocked (event E5); a warning is shown above 0.5. Default 10 µs.
Average power and energy per pulse	FR8.5: The system shall accept the average power and display the energy per pulse.	Energy per pulse = power ÷ frequency; power 10–100 W, default 100 W (supervisor).
Electrode gap	FR8.6: The system shall show the pin-to-pin electrode geometry (fixed) and allow a gap of 1, 2 or 3 mm.	Supervisor.
Pressure	FR8.7: The system shall display the pressure as 1 atm (fixed).	Supervisor.
Oxygen flow	FR8.8: The system shall accept the O_2 flow into the bubbles.	Range 100–300 sccm; default 100 sccm (supervisor).
Bubble gas	FR8.9: The system shall allow the bubble gas to be O_2 with water vapour or O_2 only.	Default O_2 + water vapour, which is required for OH, H_2O_2 and H-alpha; O_2-only mode disables Stark n_e.
EEDF	FR8.10: The system shall display the Stage 2 EEDF as Maxwellian (fixed).	Supervisor.
Waste water properties	FR8.11: The system shall accept the water volume, conductivity, initial pH, temperature, COD and waste type.	Defaults for medium-strength municipal waste water: 500 mL, 750 µS/cm, pH 7.5, 25 °C, COD 430 mg/L; volume 50–5000 mL; pH 2–12; 5–60 °C.
Target organism	FR8.12: The system shall allow the user to select the target organism (*E. coli*, *S. aureus*, bacterial spores) and enter the initial count and treatment time.	Default *E. coli*, 1.5×10^5 CFU/mL (1–2 lac, supervisor), 1 min; time 0.1–30 min; organisms without published Hom parameters are unavailable (LI-6).
Run / reset / save	FR8.13: The system shall run the disinfection simulation, reset all fields to defaults, or save the configuration on request.	Run only when all checks pass.

2.6.9 Mockup M9 – Underwater Plasma and Disinfection Results

This screen presents the Stage 2 results: disinfection performance, reactive species, survival curve, underwater OES and water chemistry. Figure 2.10 shows the mockup and Table 2.11 lists the functional requirements derived from it.

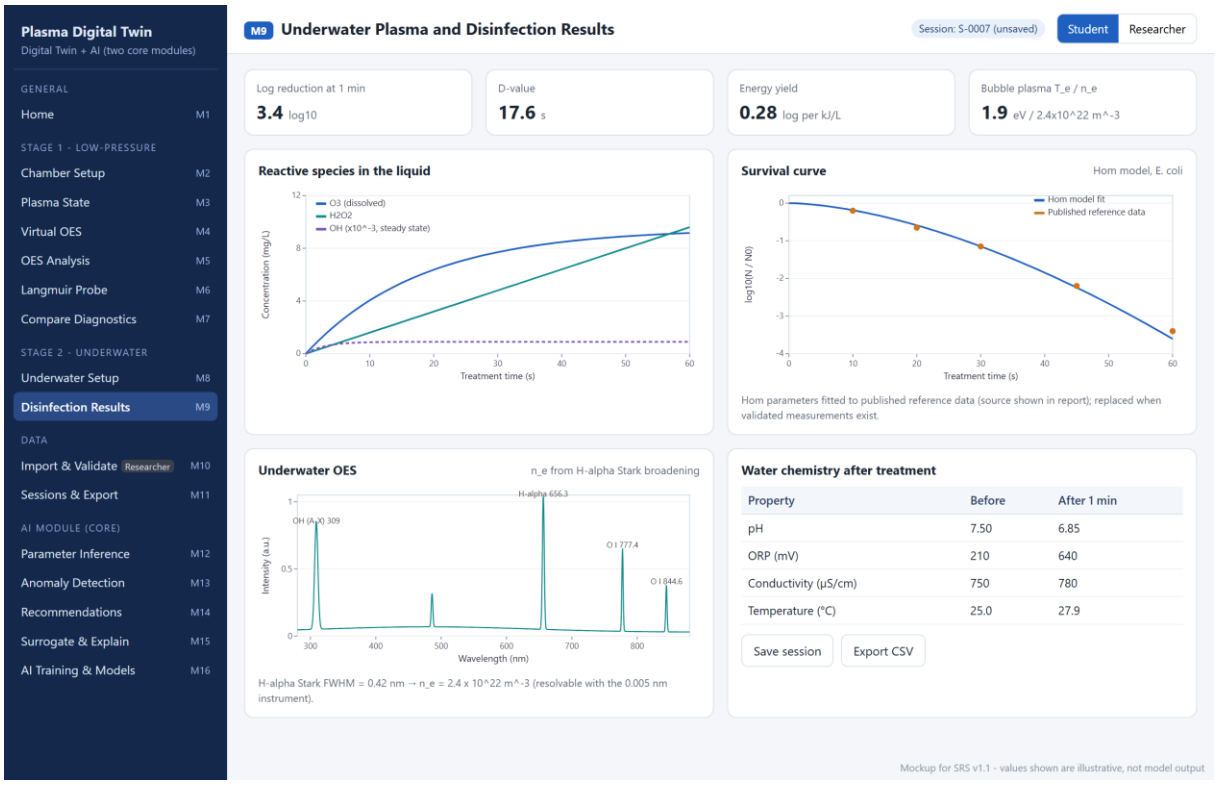


Figure 2.10: Mockup M9 – Underwater Plasma and Disinfection Results

Table 2.11: Functional requirements derived from Mockup M9 – Underwater Plasma and Disinfection Results

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Key results	FR9.1: The system shall display the log reduction at the treatment time, the D-value, the energy yield and the bubble plasma T_e and n_e.	D-value = treatment time ÷ log reduction; energy yield = log reduction ÷ (power × time ÷ volume) in log per kJ/L.
Reactive species	FR9.2: The system shall plot the concentrations of O_3, H_2O_2 and OH in the liquid against treatment time.	Gas-phase global model plus empirical transfer into the liquid (supervisor).
Survival curve	FR9.3: The system shall plot $\log_{10}(N/N_0)$ from the Hom model against time, with published reference points.	$\log_{10}(N/N_0) = -k\,C^n\,t^m$; k, n and m fitted to published time-resolved data until validated measurements exist; source shown.
Underwater OES	FR9.4: The system shall display the synthetic underwater spectrum (OH 309 nm, H-alpha 656.3 nm, O I 777 and 844 nm) and n_e from the H-alpha Stark width.	Stark n_e only when the instrument FWHM is smaller than the Stark FWHM (supervisor: H-alpha Stark method).
Water chemistry	FR9.5: The system shall display pH, ORP, conductivity and temperature before and after treatment.	Temperature rise = deposited energy ÷ (water mass × specific heat).
Save / export	FR9.6: The system shall save the Stage 2 session and export its results as CSV.	Same session rules as FR3.9.

2.6.10 Mockup M10 – Measured Data Import and Validation

This Researcher screen imports measured files, records their metadata and use, and runs literature and measured-data validation. Figure 2.11 shows the mockup and Table 2.12 lists the functional requirements derived from it.

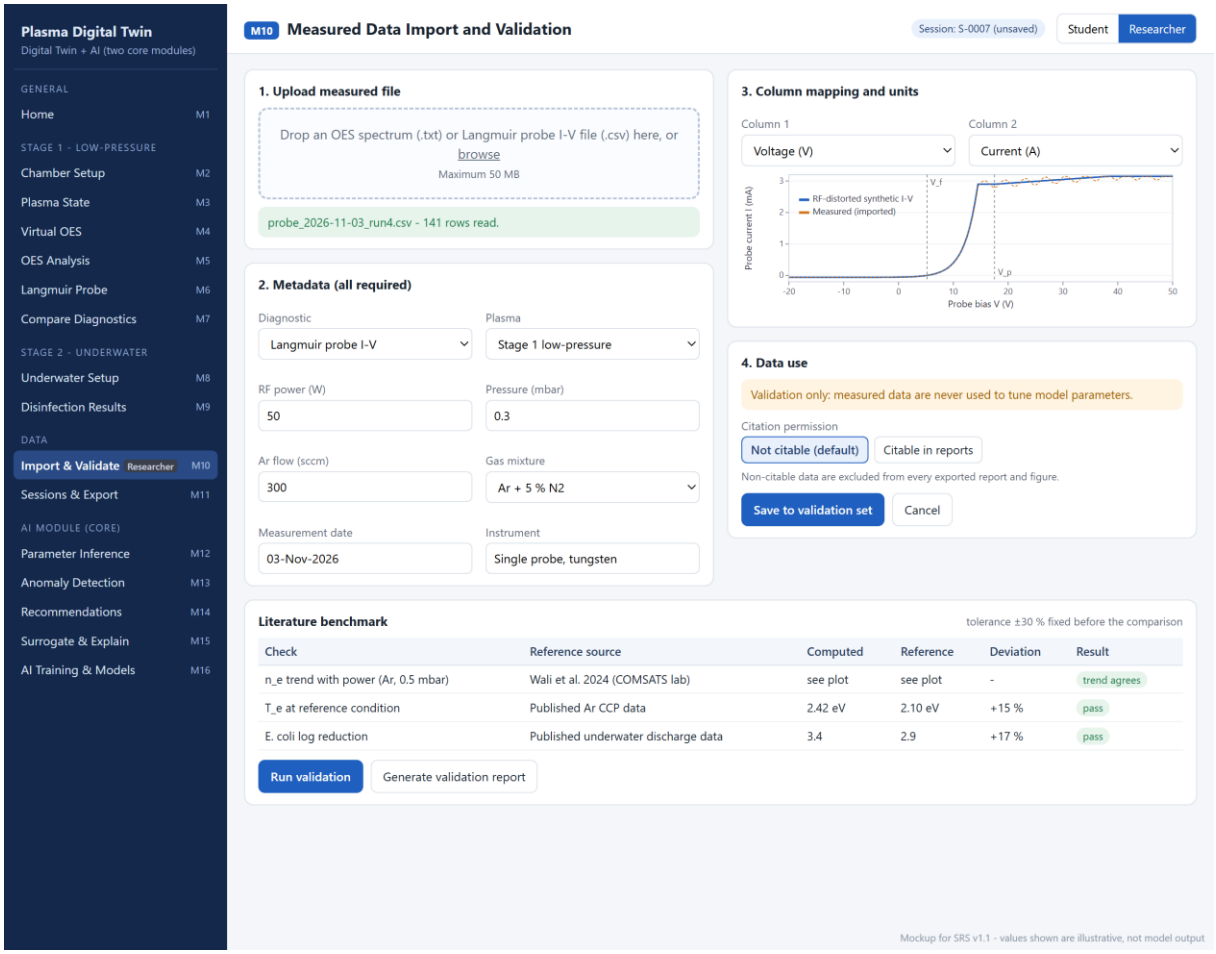


Figure 2.11: Mockup M10 – Measured Data Import and Validation

Table 2.12: Functional requirements derived from Mockup M10 – Measured Data Import and Validation

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Upload file	FR10.1: The system shall allow a Researcher to upload a measured OES spectrum (.txt) or Langmuir probe I-V curve (.csv) and report the number of rows read.	Maximum 50 MB; formats as stated by the supervisor; only visible in Researcher mode.
Metadata	FR10.2: The system shall require the diagnostic, plasma stage, operating conditions, gas mixture, measurement date and instrument for every file.	All fields mandatory; operating conditions must lie inside the envelope of the chosen stage.
Column mapping and units	FR10.3: The system shall allow the Researcher to map file columns to quantities and units and shall show a preview plot.	OES: wavelength (nm) and intensity (a.u.); probe: voltage (V) and current (A, mA or µA, converted to A).

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Data use and citation	FR10.4: The system shall state that measured data are used for validation only and shall record a citation permission for each file.	Default "Not citable" (supervisor); non-citable data are excluded from every export; measured data never tune model parameters (CON-6).
Save / cancel	FR10.5: The system shall save the file to the validation set or discard it on cancel.	Saved files are read-only.
Literature benchmark	FR10.6: The system shall list each benchmark check with its reference source, computed value, reference value, deviation and result.	Tolerance ±30 % fixed before the comparison (CON-7); reference values stored with citation.
Run validation	FR10.7: The system shall run all benchmark and measured-data comparisons on request.	Results are recorded with date and software version.
Validation report	FR10.8: The system shall generate a validation report of all checks.	Non-citable data appear only as anonymised pass/fail entries.

2.6.11 Mockup M11 – Session History and Export

This screen lists saved sessions, compares them and exports results. Figure 2.12 shows the mockup and Table 2.13 lists the functional requirements derived from it.

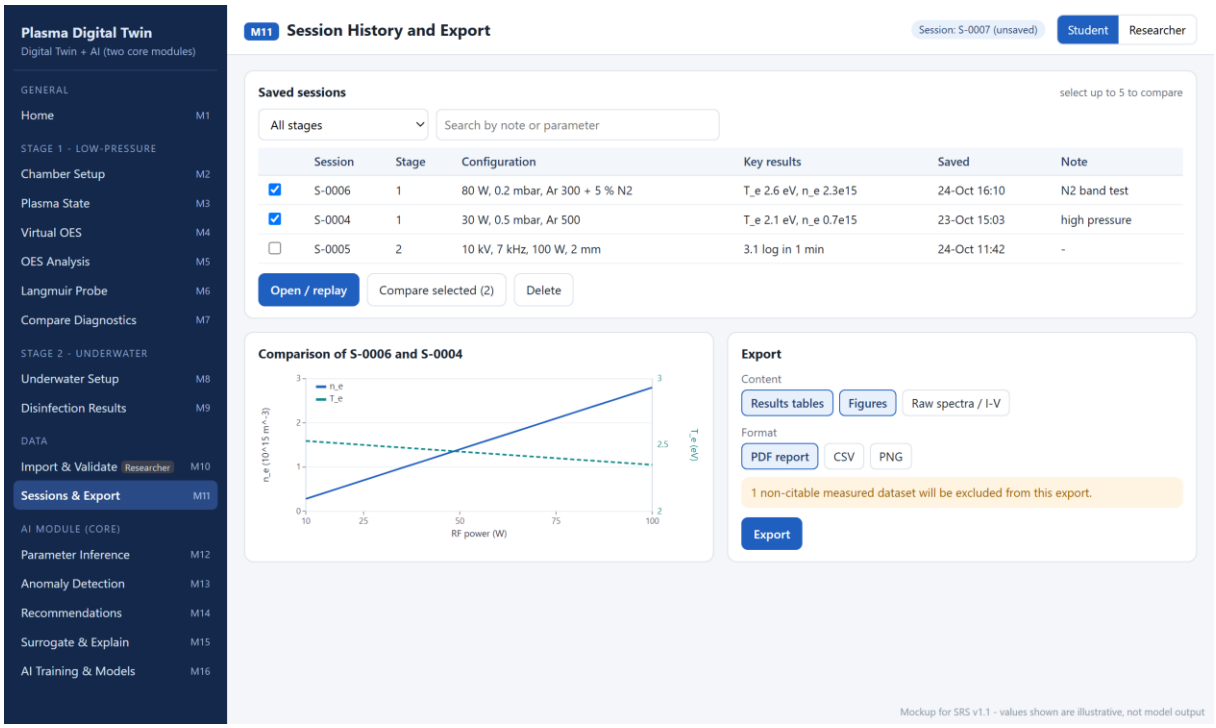


Figure 2.12: Mockup M11 – Session History and Export

Table 2.13: Functional requirements derived from Mockup M11 – Session History and Export

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Session list	FR11.1: The system shall list saved sessions with ID, stage, configuration, key results, save time and note, filtered by stage and text search.	Newest first.
Compare sessions	FR11.2: The system shall allow the user to select sessions and overlay their results in one chart.	Between 2 and 5 sessions of the same stage.
Open / replay	FR11.3: The system shall reopen a saved session, restoring its configuration and results.	If the model version has changed since saving, the user is told and offered a re-run.
Delete	FR11.4: The system shall delete selected sessions after confirmation.	Deletion is permanent and requires explicit confirmation.
Export	FR11.5: The system shall export the selected content (results tables, figures, raw spectra or I-V data) as a PDF report, CSV or PNG.	Non-citable measured data are excluded and the number excluded is shown before export (event E19).

2.6.12 Mockup M12 – AI Parameter Inference from Spectra

This AI screen (sub-module 2.1) infers plasma parameters from a spectrum with a model trained on synthetic spectra from the twin. Figure 2.13 shows the mockup and Table 2.14 lists the functional requirements derived from it.

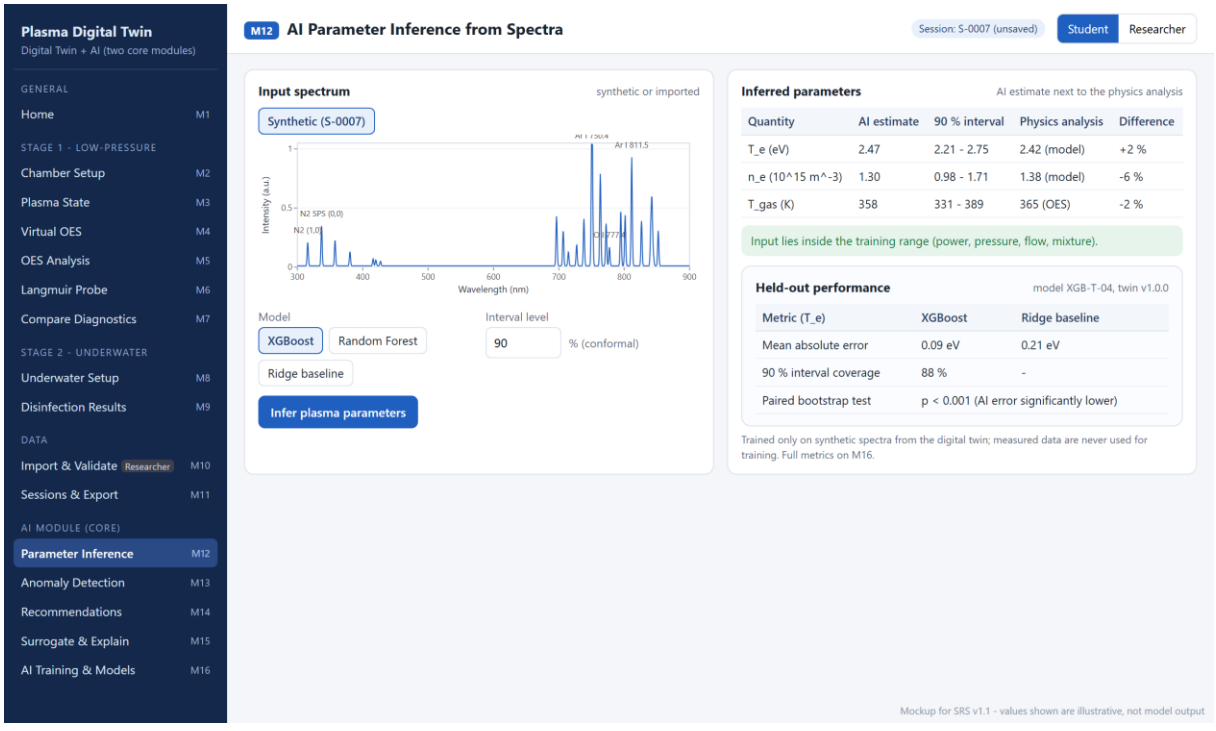


Figure 2.13: Mockup M12 – AI Parameter Inference from Spectra

Table 2.14: Functional requirements derived from Mockup M12 – AI Parameter Inference from Spectra

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Input spectrum	FR12.1: The system shall allow the user to choose the synthetic spectrum of the open session or (Researcher) an imported spectrum as input.	Imported spectra must have complete metadata (FR10.2).
Model selection	FR12.2: The system shall allow the user to choose an XGBoost model, a Random Forest model or the ridge-regression baseline.	Tree ensembles only, no deep learning (CON-10).
Infer parameters	FR12.3: The system shall estimate T_e, n_e and T_{gas} with 90 % conformal prediction intervals.	Trained only on synthetic spectra from the twin; measured data are never used for training (CON-10).
Physics comparison	FR12.4: The system shall show the physics-based value and the percentage difference next to each AI estimate.	AI never replaces the physics result (CON-12).
Training-range check	FR12.5: The system shall check that the input lies inside the training range of power, pressure, flow and mixture.	Outside the range, the estimate is withheld (event E21).
Held-out performance	FR12.6: The system shall display the held-out error of the selected model, the baseline error, the interval coverage and the result of the significance test.	Paired bootstrap of absolute errors (scipy.stats), $p < 0.05$; reported honestly whatever the outcome (AIQ-1).

2.6.13 Mockup M13 – AI Physics-Relationship Anomaly Detection

This AI screen (sub-module 2.2) flags runs and measured files whose diagnostics disagree with the physics predicted by the twin. Figure 2.14 shows the mockup and Table 2.15 lists the functional requirements derived from it.

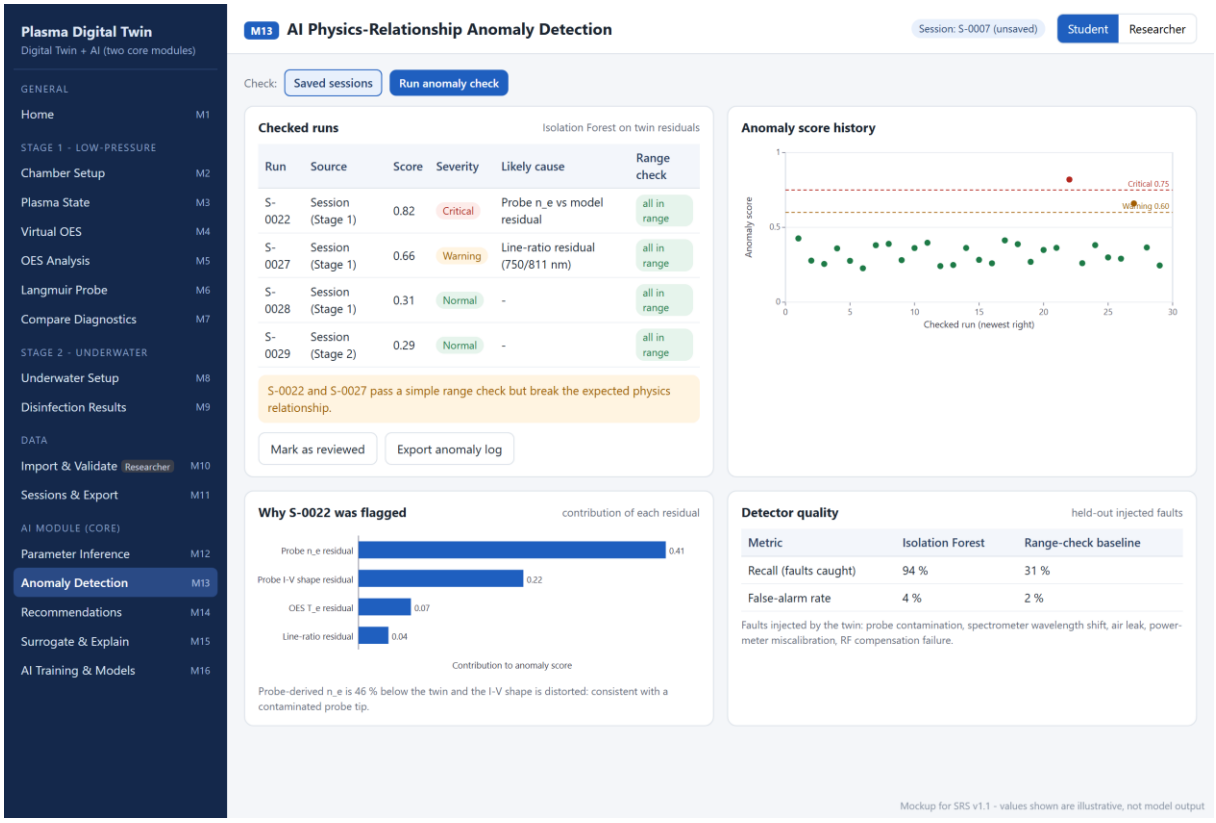


Figure 2.14: Mockup M13 – AI Physics-Relationship Anomaly Detection

Table 2.15: Functional requirements derived from Mockup M13 – AI Physics-Relationship Anomaly Detection

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Check source	FR13.1: The system shall allow the user to check saved sessions or (Researcher) imported measured files.	Imported files must have complete metadata so the twin can predict the expected diagnostics.
Run anomaly check	FR13.2: The system shall compute residuals between the twin's predictions and the observed diagnostics and score them with an Isolation Forest.	Features are physics residuals, not raw input values; a plain range check is not treated as anomaly detection.
Checked runs	FR13.3: The system shall list each checked run with its anomaly score, severity, likely cause and the result of a simple range check.	Normal < 0.60 ≤ Warning < 0.75 ≤ Critical; thresholds stored in the versioned configuration (event E24).
Score history	FR13.4: The system shall plot anomaly scores over the checked runs with the warning and critical thresholds.	—
Explanation	FR13.5: The system shall show how much each residual contributed to a flagged run's score.	The largest contribution is reported as the likely cause.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Detector quality	FR13.6: The system shall display recall and false-alarm rate on held-out injected faults for the detector and for the range-check baseline.	Faults injected by the twin: probe contamination, wavelength shift, air leak, power-meter error, RF compensation failure (AIQ-2).
Review and export	FR13.7: The system shall allow the user to mark a flagged run as reviewed and to export the anomaly log.	The log is stored in SQLite with time stamps and session IDs.

2.6.14 Mockup M14 – AI Recommendations (Inverse Design)

This AI screen (sub-module 2.4) proposes settings for a user goal and verifies every proposal with the digital twin. Figure 2.15 shows the mockup and Table 2.16 lists the functional requirements derived from it.

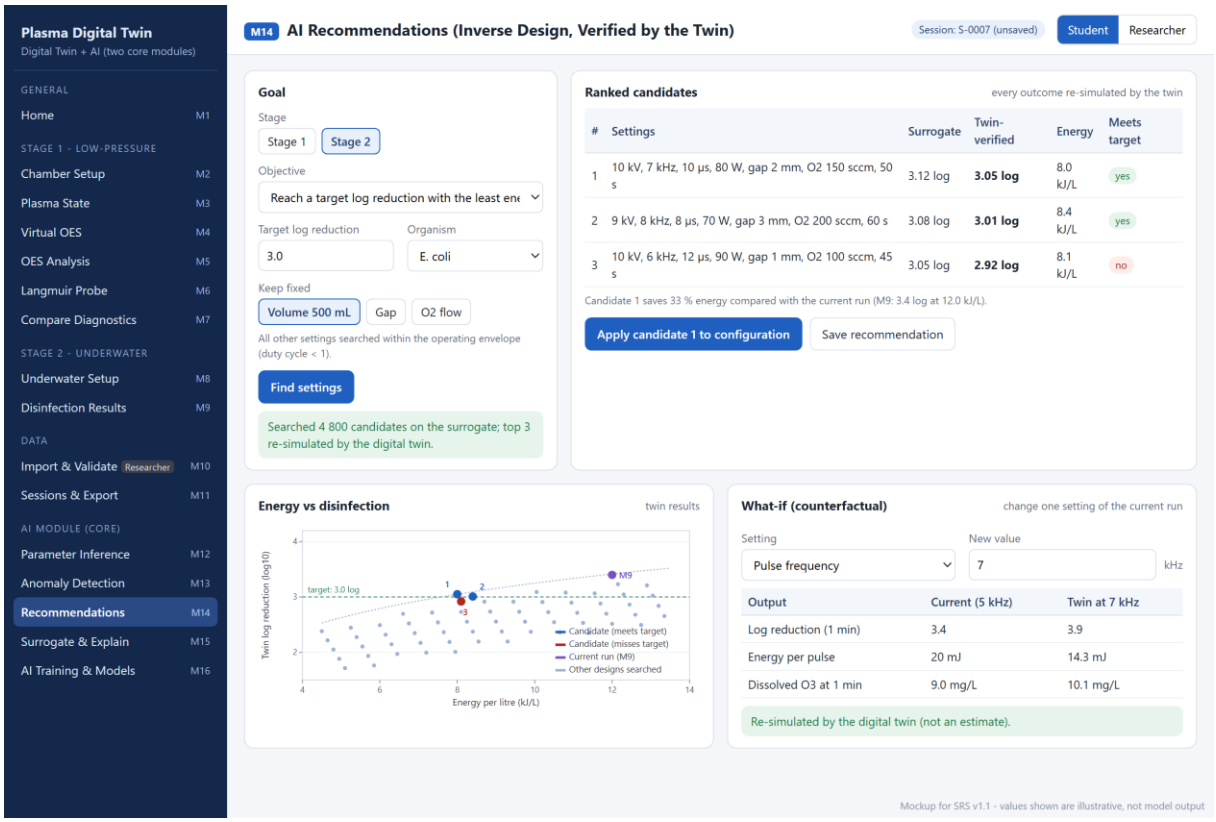


Figure 2.15: Mockup M14 – AI Recommendations (Inverse Design)

Table 2.16: Functional requirements derived from Mockup M14 – AI Recommendations (Inverse Design)

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Goal	FR14.1: The system shall allow the user to choose the stage, an objective and a target value.	Stage 2: target log reduction with least energy, or maximum log reduction within an energy budget; Stage 1: target n_e, target T_{gas} or strongest chosen line.
Fixed settings	FR14.2: The system shall allow the user to keep chosen settings fixed during the search.	All other settings are searched only inside the operating envelope, with duty cycle < 1.
Find settings	FR14.3: The system shall search the envelope on the surrogate and re-simulate the best three candidates with the digital twin (event E25).	No recommendation is shown without its twin re-simulation (CON-12).
Ranked candidates	FR14.4: The system shall list each candidate's settings, surrogate prediction, twin-verified outcome, energy per litre and whether the goal is met.	Candidates are ranked by the twin-verified outcome, not by the surrogate.
Energy chart	FR14.5: The system shall plot the twin outcomes of the candidates and the current run against energy per litre.	—
What-if	FR14.6: The system shall let the user change one setting of the current run and show the twin's re-simulated outputs.	Values come from the twin, not from an estimate.
Apply or save	FR14.7: The system shall apply a chosen candidate to the configuration screen or save the recommendation.	Applying fills M2 or M8; nothing runs until the user presses Run.
Unreachable goal	FR14.8: The system shall state when no setting inside the envelope reaches the goal and show the closest twin-verified option.	Event E26.

2.6.15 Mockup M15 – AI Surrogate Model and Explainability

This AI screen (sub-module 2.3) uses a fast surrogate of the twin for large explorations and explains its outputs with SHAP. Figure 2.16 shows the mockup and Table 2.17 lists the functional requirements derived from it.

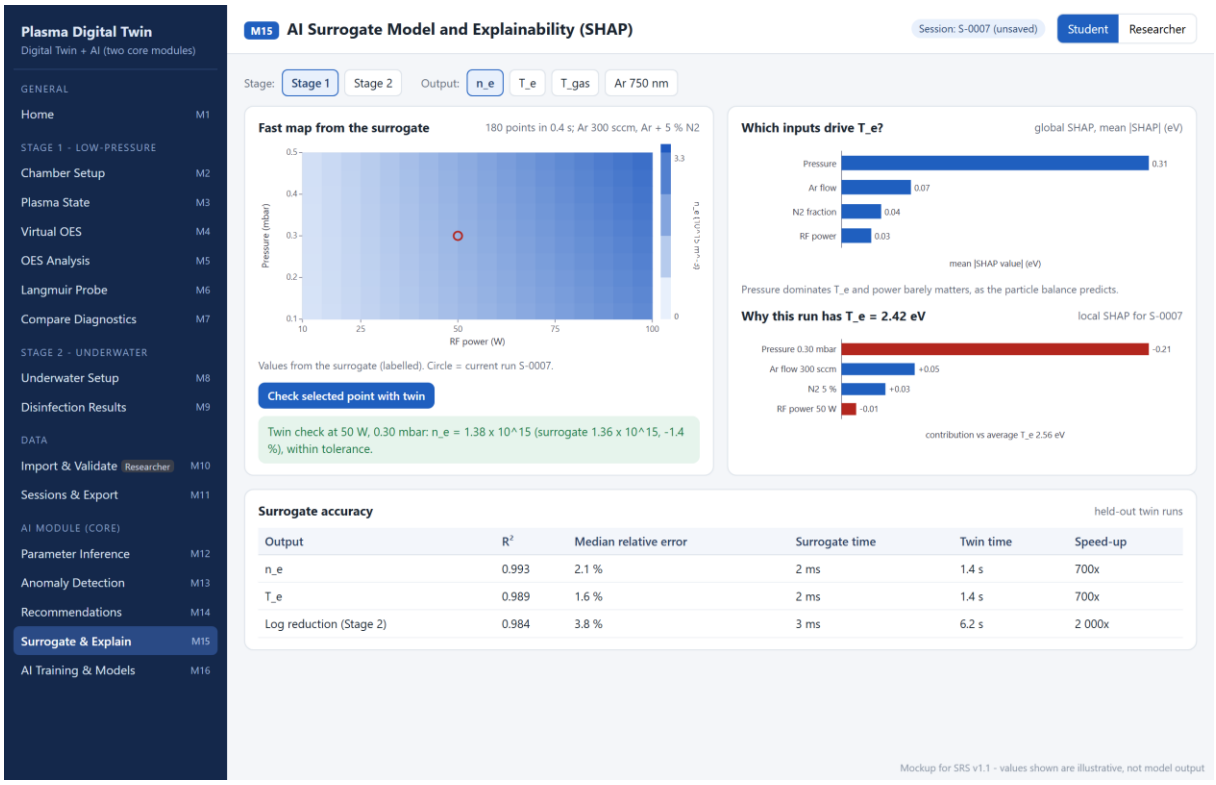


Figure 2.16: Mockup M15 – AI Surrogate Model and Explainability

Table 2.17: Functional requirements derived from Mockup M15 – AI Surrogate Model and Explainability

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Stage and output	FR15.1: The system shall allow the user to choose the stage and the output to explore.	—
Fast map	FR15.2: The system shall draw a map of the chosen output over two inputs using the surrogate.	Every surrogate value is labelled "surrogate" (CON-12).
Check with twin	FR15.3: The system shall re-run a selected point through the twin and show both values.	If they differ by more than 5 %, the twin value is shown and the region is flagged (event E27).
Global explanation	FR15.4: The system shall show global SHAP importance of each input for the chosen output.	—
Local explanation	FR15.5: The system shall show the SHAP contribution of each input for the current run.	Local SHAP values add up to the prediction minus the average prediction.
Surrogate accuracy	FR15.6: The system shall display R^2, median relative error and speed-up against the twin for each output.	Acceptance: $R^2 \geq 0.98$, median relative error ≤ 5 %, at least 100 times faster (AIQ-3).

2.6.16 Mockup M16 – AI Training, Evaluation and Model Registry

This AI screen (sub-module 2.5) shows the training data, trains models (Researcher) and lists every registered model. Figure 2.17 shows the mockup and Table 2.18 lists the functional requirements derived from it.

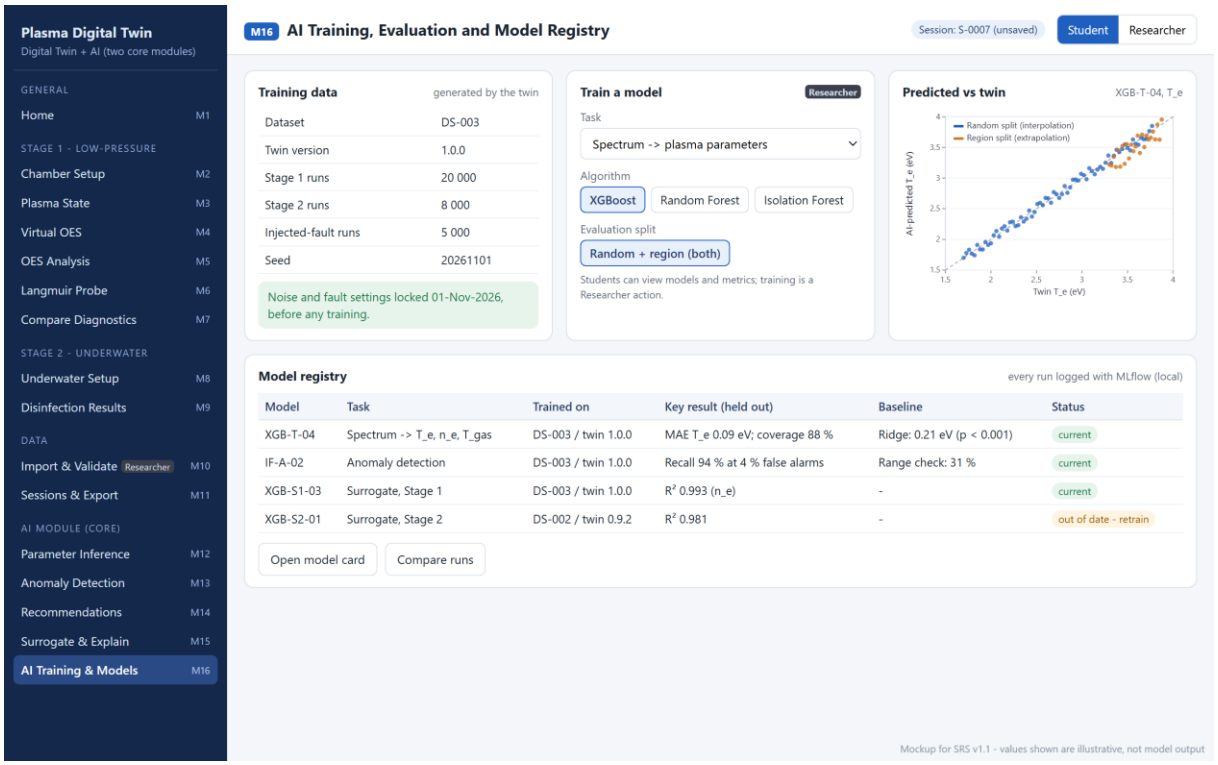


Figure 2.17: Mockup M16 – AI Training, Evaluation and Model Registry

Table 2.18: Functional requirements derived from Mockup M16 – AI Training, Evaluation and Model Registry

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Training data	FR16.1: The system shall display the dataset ID, twin version, number of runs per stage and fault type, and the seed.	Noise and fault settings are locked before any training; the lock date is shown (AIQ-5).
Train a model	FR16.2: The system shall allow a Researcher to train a model by choosing the task, the algorithm and the evaluation split.	Researcher only; Students can view models and metrics (event E23).
Evaluation	FR16.3: The system shall evaluate every model on a random split and a region-based split and plot predicted against twin values.	Both results are always reported; the region split tests extrapolation.
Model registry	FR16.4: The system shall list every model with its task, dataset, twin version, key held-out result, baseline result and status.	A model stamped with an older twin version is marked out of date and is not used (event E22, AIQ-6).
Model card and runs	FR16.5: The system shall open a model's card and compare logged training runs.	Every run is logged with MLflow in a local file store (SI-7).

2.6.17 Backend Event–Response Table

The following events capture backend processing that is not visible as a screen element. Events E1–E27 were identified for the Stage 1 simulation, the diagnostics, Stage 2, data import, sessions, export and the AI module. Table 2.19 records the event list, and Table 2.20 records the trigger, source, states, response and exception handling of each event.

Table 2.19: Identified backend events

ID	Event	ID	Event
E1	Application start-up	E15	Validation run requested
E2	Stage 1 simulation requested	E16	Validation tolerance exceeded
E3	Configuration outside envelope	E17	Parameter sweep completed
E4	Solver non-convergence	E18	Session save failure
E5	Inconsistent Stage 2 pulse settings	E19	Export with non-citable data
E6	Reference data missing or corrupt	E20	Stage 2 disinfection simulation requested
E7	Synthetic spectrum requested	E21	AI input outside training range
E8	Doppler method below instrument resolution	E22	AI model missing or out of date
E9	Insufficient Boltzmann-plot lines	E23	AI training run requested
E10	Probe I-V generation requested	E24	Anomaly detected
E11	Probe analysis failure	E25	Recommendation requested
E12	OML validity violated	E26	Recommendation target unreachable
E13	Measured file uploaded	E27	Surrogate–twin disagreement
E14	File rejected (format, size or parse failure)		

Table 2.20: Event–response table for backend processes

Event Name	Trigger	Source	System State (Before)	Response	System State (After)	Exception Handling
E1 Application start-up	User starts the application	Student / Researcher (launcher)	Not running	1. Load and checksum reference data files 2. Open or create the SQLite database and apply schema migrations 3. Show Home with data status	Home displayed; ready	Missing file → E6; unreadable database → back it up, create a new one, notify user

Event Name	Trigger	Source	System State (Before)	Response	System State (After)	Exception Handling
E2 Stage 1 simulation requested	Run simulation (M2) or quick start (M1)	Student	Valid configuration	1. Compute configuration hash; return cached result if present 2. Compute rate coefficients for the chosen EEDF 3. Solve particle balance for T_e (bracketed root finding) 4. Solve power balance for n_e and energy balance for T_{gas} 5. Compute species, sheath and 1D profile if selected	Results shown in M3; unsaved session	Non-convergence → E4
E3 Configuration outside envelope	Input changed to an out-of-range or inconsistent value	Configuration Validator	Editing	1. Highlight the field 2. Show allowed range and unit 3. Disable Run	Configuration invalid; Run disabled	None needed; restored when corrected
E4 Solver non-convergence	Root not bracketed or iteration limit reached	Simulation Orchestrator	Computing	1. Retry once with a widened bracket 2. If still failing, stop and log inputs and residual 3. Show message with nearest valid configuration	Previous results kept; no partial results shown	Logged for regression testing (REL-1)
E5 Inconsistent Stage 2 pulse settings	Duty cycle ≥ 1 (pulse width × frequency)	Configuration Validator	Editing M8	1. Block Run 2. Show computed duty cycle and corrective action	Configuration invalid	Warning only when $0.5 < \text{duty} < 1$
E6 Reference data missing or corrupt	File absent or checksum mismatch	Reference Data Source	Start-up or on demand	1. Mark data set "missing" in M1 2. Disable dependent features (e.g., OES without line data)	Degraded mode	Instruct user to reinstall the data package
E7 Synthetic spectrum requested	Send to OES (M3) or Recompute (M4)	Student	Solved plasma state available	1. Solve CR model for 1s and 2p populations 2. Compute line intensities from NIST data 3. Convolve with instrument function; add seeded noise	Spectrum and line table shown	No plasma state → prompt to run a simulation first
E8 Doppler below resolution	Doppler method selected and instrument FWHM > expected Doppler FWHM	OES Analysis	Analysing	1. Disable Doppler option with explanation 2. Offer N_2 rotational method	Alternative method used	If no N_2 in mixture, T_{gas} reported from model only, marked
E9 Insufficient Boltzmann lines	< 4 usable lines or energy span < 1.0 eV	OES Analysis	Analysing	1. Do not report T_{exc} 2. Show which condition failed	T_{exc} unavailable	User may add lines or widen range
E10 Probe I-V requested	Generate I-V curve (M6)	Student	Solved plasma state available	1. Compute ion and electron currents (OML) 2. Apply RF averaging if enabled; add seeded noise 3. Run analysis automatically	Curve and results shown	No plasma state → prompt to run a simulation

Event Name	Trigger	Source	System State (Before)	Response	System State (After)	Exception Handling
E11 Probe analysis failure	V_f or V_p cannot be found (no zero crossing)	Probe Analysis	Analysing	1. Keep the raw curve 2. Report the failed step 3. Suggest a wider sweep	Partial results only	Logged
E12 OML validity violated	$r_p/\lambda_D > 1$	Probe Analysis	Analysis complete	1. Show warning 2. Mark n_e as approximate	Results flagged	None
E13 Measured file uploaded	Researcher drops or browses a file (M10)	Researcher (from OES Spectrometer or Probe System)	Researcher mode	1. Check type and size 2. Detect delimiter; parse; show rows read and preview 3. Request metadata	File pending metadata	Failure → E14
E14 File rejected	Wrong type, > 50 MB, or unparsable	Import & Parsing	Upload in progress	1. Reject file 2. Show exact reason (line number if parse error)	Nothing stored	None
E15 Validation run requested	Run validation (M10)	Researcher	Benchmarks and/or measured files present	1. Load tolerance (±30 %) from versioned file 2. Compute each deviation 3. Record pass/fail with date and version	Validation results stored	Missing reference → check skipped and listed
E16 Tolerance exceeded	|deviation| > 30 %	Validation & Benchmark	Validation running	1. Mark check as fail 2. Keep the result in the report	Fail recorded	Parameters are never adjusted automatically (CON-6)
E17 Parameter sweep completed	Last sweep point solved	Simulation Orchestrator	Sweep running with progress	1. Store series 2. Plot results in M3	Sweep results available	Failed points listed, others kept
E18 Session save failure	Database locked or disk full	Session Manager	Save requested	1. Retry once after 1 s 2. Keep results in memory 3. Offer CSV export	Session unsaved; results kept	User notified with cause
E19 Export with non-citable data	Export requested while selection includes non-citable data	Export & Report	Export dialog	1. Remove non-citable datasets 2. Show how many were excluded 3. Add an exclusion note	Export file created without them	None
E20 Stage 2 simulation requested	Run disinfection simulation (M8)	Student	Valid Stage 2 configuration	1. Solve bubble global model (Maxwellian) 2. Transfer species to liquid 3. Integrate Hom survival and water chemistry 4. Synthesise underwater OES and Stark n_e	Results shown in M9	Non-convergence → E4 handling
E21 AI input outside training range	Input conditions outside training data	AI Inference	Inference requested	1. Withhold the estimate 2. Show the out-of-range parameters	Physics result only	None
E22 AI model missing or out of date	Model file missing, or model stamped with an older twin version	AI Training & Registry	Start-up or twin version changed	1. Mark the affected AI screens "retrain required" 2. Never use an out-of-date model for a result 3. Offer training (Researcher)	That AI task disabled until retrained	Physics results unaffected

Event Name	Trigger	Source	System State (Before)	Response	System State (After)	Exception Handling
E23 AI training run requested	Start training (M16)	Researcher	Dataset settings locked	1. Generate or load the twin dataset with the recorded seed 2. Train the model and its baseline 3. Evaluate on random and region splits 4. Log the run in MLflow; write the model card	New model registered with twin version	Training failure → previous model kept; error logged
E24 Anomaly detected	Anomaly score ≥ warning threshold	AI Anomaly Detector	Anomaly check running	1. Assign Warning or Critical severity 2. Name the largest contributing residual 3. Add the run to the anomaly log	Run flagged for review	None
E25 Recommendation requested	Find settings (M14)	Student	Valid goal and constraints	1. Search the envelope on the surrogate 2. Re-simulate the top 3 candidates with the twin 3. Rank by twin-verified outcome	Ranked candidates shown	Twin fails for a candidate → candidate dropped and listed
E26 Recommendation target unreachable	No twin-verified candidate meets the goal	Recommendation Engine	Search complete	1. State that the goal cannot be reached inside the envelope 2. Show the closest twin-verified candidate	Closest option shown	None
E27 Surrogate–twin disagreement	|surrogate − twin| > 5 % at a checked point	AI Surrogate	Twin check requested	1. Show the twin value 2. Flag the region as unreliable 3. Log the point for retraining	Twin value used	None

2.6.18 Functional Requirements Derived from Events

Table 2.21 lists the functional requirements derived from the event–response table.

Table 2.21: Functional requirements derived from the event–response table

Event	Functional Requirement (FR-ID: Statement)	Business Rule
E1	FR17.1: The system shall verify the presence and checksum of every reference data file and open or create the session database at start-up.	Checksums are stored in a manifest shipped with the data.
E2	FR17.2: The system shall compute the Stage 1 plasma state by solving the particle balance for T_e and then the power balance for n_e, reusing a cached result for an identical configuration.	Cache key = hash of all inputs and the model version.
E3	FR17.3: The system shall block a run while any input is outside its allowed range and shall state the allowed range.	Ranges come from one configuration file shared by UI and validator.
E4	FR17.4: The system shall retry a non-converged solve once with a widened bracket and otherwise report failure without showing partial results.	Convergence residual < 10^{-8}.
E5	FR17.5: The system shall block a Stage 2 run whose duty cycle is 1 or more.	Warning when duty cycle > 0.5.
E6	FR17.6: The system shall disable features whose reference data are missing or corrupt and show this on the Home screen.	No silent fallback to built-in guesses.
E7	FR17.7: The system shall synthesise spectra from CR-model populations and NIST line data with a seeded instrument noise model.	Same seed and inputs give the same spectrum.
E8	FR17.8: The system shall disable Doppler gas-temperature analysis when the instrument FWHM exceeds the expected Doppler FWHM.	Doppler FWHM computed for argon at the model T_{gas}.
E9	FR17.9: The system shall not report T_{exc} unless at least 4 lines spanning at least 1.0 eV are used.	—
E10, E11	FR17.10: The system shall generate and automatically analyse probe curves and report the step at which an analysis fails.	Raw curve is always kept.
E12	FR17.11: The system shall flag probe-derived n_e as approximate when $r_p/\lambda_D > 1$.	—
E13, E14	FR17.12: The system shall parse uploaded files and reject wrong-type, oversized or unparsable files with the exact reason.	Maximum 50 MB; nothing stored on rejection.
E15, E16	FR17.13: The system shall evaluate every validation check against the ±30 % tolerance loaded from the versioned file and record failures without changing model parameters.	Tolerance fixed before comparison (CON-7).
E17	FR17.14: The system shall show sweep progress and keep all solved points when some points fail.	Progress updated at least every 2 s (PER-4).
E18	FR17.15: The system shall keep results in memory and offer CSV export when a session cannot be saved.	One automatic retry.

Event	Functional Requirement (FR-ID: Statement)	Business Rule
E19	FR17.16: The system shall exclude non-citable measured data from every export and report the number excluded.	Citation permission recorded per file (FR10.4).
E20	FR17.17: The system shall compute the Stage 2 bubble plasma, liquid species, Hom survival, water chemistry and underwater spectrum in one run.	Maxwellian EEDF (CON-4).
E21, E22	FR17.18: The system shall withhold AI estimates outside the training range and shall not use any AI model that is missing or trained on an older twin version.	AI outputs never come from an out-of-date model (AIQ-6).
E23	FR17.19: The system shall train each AI model together with its baseline on twin-generated data, evaluate both on random and region splits, and log the run with MLflow.	Dataset settings and seed fixed before training (CON-10).
E24	FR17.20: The system shall flag runs whose anomaly score reaches the warning threshold and record the likely cause.	Thresholds come from the versioned configuration.
E25, E26	FR17.21: The system shall verify every recommendation candidate with the twin and report when a goal cannot be reached inside the envelope.	Counterfactual principle: no unverified advice (CON-12).
E27	FR17.22: The system shall prefer the twin value whenever the surrogate differs from it by more than 5 %.	The twin is the reference; the surrogate is only a fast approximation.

2.7 Non-Functional Requirements

The non-functional requirements below are written in measurable terms following Wiegers and Beatty [33]. Each table gives the target, the reason the target was chosen, and how the architecture in Chapter 3 supports it. Performance targets refer to a reference computer with a 4-core 2.0 GHz or faster CPU and 8 GB RAM.

2.7.1 Reliability

Table 2.22 lists the reliability requirements. A failure is defined as a crash, an unhandled exception shown to the user, a lost saved session or a result that differs from the reference regression value.

Table 2.22: Reliability requirements

ID	Requirement (measurable)	Justification	Architectural support
REL-1	The Stage 1 solver shall converge for 100 % of a pre-registered grid of at least 500 configurations spanning the full envelope.	A non-converging run blocks learning; the grid exposes edge cases before users do.	Bracketed root finding with retry (E4); grid test in pytest CI.
REL-2	No unhandled exception shall reach the user for any of the 27 events; every exception path in the event–response table shall produce a message.	Students must always see a cause and a corrective action.	Central error handler in the application-services layer; one test per event.

ID	Requirement (measurable)	Justification	Architectural support
REL-3	Zero saved sessions shall be lost in 50 forced-termination tests during saving.	Sessions are the user's experimental record.	SQLite transactions with write-ahead logging.
REL-4	Re-running a saved configuration with the same seed and model version shall reproduce every numeric result to a relative difference below 10^{-9}.	Scientific results must be reproducible for defence and validation.	Seeded noise; model version stored with each session.

2.7.2 Usability

Table 2.23 lists the usability requirements for a learner with little software experience.

Table 2.23: Usability requirements

ID	Requirement (measurable)	Justification	Architectural support
USE-1	A first-time user shall run the default Stage 1 experiment from the Home screen with no more than 2 interactions.	Immediate success motivates learning.	Quick-start list on M1 calls the same run service as M2.
USE-2	At least 80 % of at least 5 student test users shall configure a stated Stage 1 run and read T_{gas} within 5 minutes without help.	Directly supports BO-2.	Stage-grouped navigation; defaults from the laboratory set-up.
USE-3	100 % of input fields shall show the allowed range and unit, and 100 % of error messages shall state the cause and a corrective action.	Prevents invalid runs and frustration.	Ranges and messages generated from one configuration file.
USE-4	Every plotted axis shall show a symbol and SI unit, and the glossary shall cover 100 % of displayed symbols.	Physics results are meaningless without units.	Shared chart helper enforces axis labels.
USE-5	Status shall never be conveyed by colour alone, and text contrast shall be at least 4.5:1.	WCAG 2.1 AA accessibility [34].	Status component always renders a text label.

2.7.3 Performance

Table 2.24 lists the performance requirements. The targets follow Nielsen's response-time limits: about 1 s keeps the user's flow of thought and 10 s is the limit for keeping attention without a progress indicator [35].

Table 2.24: Performance requirements

ID	Requirement (measurable)	Justification	Architectural support
PER-1	95 % of Stage 1 0D runs shall complete within 2 s, and 0D + 1D runs within 5 s.	Close to the 1 s limit so students can explore parameters interactively.	Cached results keyed on configuration hash; vectorised NumPy; compiled SciPy root finding.

ID	Requirement (measurable)	Justification	Architectural support
PER-2	A synthetic spectrum for 300–900 nm at 0.005 nm resolution (120 000 points) shall be recomputed within 3 s.	Instrument settings are changed repeatedly while learning.	Instrument convolution by FFT; plasma state not re-solved (FR4.4).
PER-3	Probe I-V generation and analysis for up to 2001 points shall complete within 2 s.	Same interactive limit as PER-1.	Analytic OML currents; vectorised smoothing and differentiation.
PER-4	A 200-point sweep shall complete within 120 s, with progress updated at least every 2 s.	Beyond 10 s a progress indicator is required.	Each point reuses the cached rate tables; progress callback.
PER-5	A Stage 2 disinfection run for a 1 min treatment shall complete within 10 s.	Within the attention limit.	Stiff ODE solver for species kinetics.
PER-6	Validation feedback after an input change shall appear within 0.5 s.	Immediate feedback prevents invalid runs.	Validation runs before, and separately from, the model.
PER-7	A 50 MB file shall be parsed and previewed within 15 s.	Largest allowed file.	Streaming CSV parser; preview of a downsampled curve.
PER-8	The Home screen shall be ready within 15 s of starting the application.	Users judge the tool at first launch.	Reference data loaded once and cached.
PER-9	One AI parameter inference shall complete within 1 s.	Within the 1 s flow limit.	Trained model loaded once and cached.
PER-10	A recommendation, including twin re-simulation of three candidates, shall complete within 60 s, with progress shown.	Long search; progress indicator required beyond 10 s.	Search runs on the fast surrogate; only the top candidates use the twin.

2.7.4 Security

Table 2.25 lists the security requirements. The system is a single-user local application, so the main assets are the user's measured data and the integrity of results.

Table 2.25: Security requirements

ID	Requirement (measurable)	Justification	Architectural support
SEC-1	The application shall accept connections only on the loopback interface; a port scan from another computer shall find no open application port.	Measured laboratory data must not be exposed on the network.	Server bound to 127.0.0.1.
SEC-2	100 % of uploaded files shall be treated as data only: no content is executed, and files of the wrong type or over 50 MB are rejected.	Imported files come from other computers.	Import & Parsing service with type, size and schema checks (E14).
SEC-3	Non-citable measured data shall appear in 0 exported files, figures or reports in automated export tests.	Supervisor forbids citing the data (LI-7).	Citation filter inside the Export & Report service (E19).
SEC-4	100 % of database statements shall use parameterised queries; injection test strings in notes and search fields shall have no effect.	Free-text fields are stored in SQLite.	Session Manager data-access functions.

ID	Requirement (measurable)	Justification	Architectural support
SEC-5	The application shall make no outbound network connection at run time.	Works offline; no data leave the computer.	Reference data shipped locally (OE-4).

2.7.5 Accuracy and Scientific Validity

Table 2.26 lists the requirements that make the results scientifically defensible.

Table 2.26: Accuracy and scientific validity requirements

ID	Requirement (measurable)	Justification	Architectural support
ACC-1	Model T_e, n_e and T_{gas} shall agree with published reference values within ±30 % at no fewer than 3 Stage 1 operating points, and the log reduction at no fewer than 1 Stage 2 point.	Supervisor-accepted tolerance; semi-quantitative validation (BO-1, BO-4).	Validation & Benchmark service with versioned reference values.
ACC-2	Analysis of a noise-free synthetic probe curve shall recover the model T_e and n_e within ±10 %; OES line-ratio T_e shall agree within ±30 %.	A virtual instrument must return what was put in; this tests the analysis code.	Domain modules tested in isolation in pytest.
ACC-3	Power-balance shares shall sum to 100 ± 0.5 % and the particle-balance residual shall be below 10^{-8} for every accepted run.	Energy and particle conservation are physical laws.	Checks inside the Simulation Orchestrator before results are accepted.
ACC-4	Every reference value and the tolerance shall be fixed in a versioned file before a comparison is run; 0 changes are allowed after results are seen without a new version entry.	Prevents tuning to the answer.	Reference data store under version control.

2.7.6 AI Model Quality

Table 2.27 lists the quality requirements for the AI module. They make the AI results measurable and honest: every model is compared with a simple baseline, tested on data it has not seen, and checked against the twin.

Table 2.27: AI model-quality requirements

ID	Requirement (measurable)	Justification	Architectural support
AIQ-1	Spectrum inference shall have a significantly lower error than the ridge baseline (paired bootstrap, $p < 0.05$) and 90 % interval coverage ≥ 85 % on held-out synthetic spectra, reported for random and region splits.	BO-6; a model must beat a simple alternative to justify itself.	Registry stores baseline and model results side by side.
AIQ-2	Anomaly detection shall reach recall ≥ 90 % at a false-alarm rate ≤ 5 % on held-out injected faults, with recall significantly above the range-check baseline.	BO-7; detects faults a range check cannot see.	Fault injection in the twin dataset generator.

ID	Requirement (measurable)	Justification	Architectural support
AIQ-3	Surrogates shall reach $R^2 \geq 0.98$ and median relative error ≤ 5 % on held-out twin runs and be at least 100 times faster than the twin.	BO-8; fast exploration without misleading values.	Twin check on demand (E27).
AIQ-4	100 % of recommendations shall carry a twin-verified outcome, and the twin shall confirm the goal for ≥ 90 % of a benchmark set of goals.	BO-9; advice must be quantified and verified.	Recommendation engine must call the twin before returning.
AIQ-5	Dataset noise, fault settings and seed shall be fixed before training, every training run shall be logged, and 0 models shall be trained on measured data.	Prevents tuning the data until the numbers look good.	Locked dataset configuration; MLflow registry.
AIQ-6	0 AI results shall be produced by a model stamped with a twin version other than the current one.	AI must stay consistent with the physics it learned from.	Twin-version stamp checked at load (E22).

2.7.7 Maintainability and Portability

Table 2.28 lists the maintainability and portability requirements.

Table 2.28: Maintainability and portability requirements

ID	Requirement (measurable)	Justification	Architectural support
MNT-1	Each domain module shall have its own pytest file, and line coverage of the domain layer shall be at least 80 %.	Students must defend and change the code safely.	Layered design isolates domain code from the UI.
MNT-2	100 % of physical constants and reference data shall be stored in data files with a citation, not hard-coded in functions.	Every number must be traceable to a source.	Reference data store (data layer).
MNT-3	100 % of public functions shall have type hints and a docstring that explains the physics or statistics reason.	Code is defended live before the committee.	Checked in CI.
POR-1	The system shall install from one pinned requirements file and start with one command on Windows 10/11 and Ubuntu 22.04+.	Laboratory and student computers differ.	Pure-Python stack; no compiled project code.

2.8 External Interface Requirements

2.8.1 User Interfaces Requirements

The mockups in Section 2.6 communicate the intended interaction; they are not the committed screen designs. The user interface shall follow these conventions:

- A left navigation panel groups screens as General, Stage 1 – Low-Pressure, Stage 2 – Underwater, Data and AI Module (Core), and shows each screen's mockup ID; Researcher-only items carry a "Researcher" badge.

- AI and surrogate values are always labelled as such and shown next to the physics result.
- A role switch (Student / Researcher) and the current session ID are visible on every screen.
- Fonts are sans-serif at no less than 12 px for labels; every numeric field shows its unit and allowed range.
- Messages use three levels with an icon and text: success (green), warning (amber) and blocking error (red).
- Primary actions (Run, Generate, Analyse, Export) appear at the bottom of each screen as filled buttons.
- Charts are interactive (zoom, hover values, reset by double-click, PNG download).
- Layouts work from 1366 × 768 upwards; all controls are reachable with the keyboard in a logical tab order.
- Help and the glossary are reachable from the Home screen; every symbol links to its glossary entry.

2.8.2 Software Interfaces

SI-1: NIST Atomic Spectra Database. The system shall read an installed snapshot of argon, oxygen and hydrogen line data (wavelength, A_{ki}, g_k, E_k) in CSV form, with the retrieval date recorded.

SI-2: LXCat cross-sections. The system shall read electron–argon, nitrogen and oxygen cross-section files in LXCat text format and integrate them over the selected EEDF to obtain rate coefficients.

SI-3: SQLite 3. The system shall store sessions, configurations, imported-file metadata and validation results through Python's sqlite3 interface.

SI-4: Measured OES files. The system shall read text files with two numeric columns (wavelength in nm and intensity) separated by spaces, tabs or commas; lines starting with # are treated as comments.

SI-5: Measured probe files. The system shall read .csv files with voltage and current columns, with a header row optional.

SI-6: Exports. The system shall write PDF reports, UTF-8 comma-separated CSV files and PNG figures at 300 dpi.

SI-7: MLflow (local). The system shall log AI training runs (parameters, metrics, model files) to a local MLflow file store through the classic tracking API; no tracking server is used.

2.8.3 Hardware Interfaces

HI-1: The system has no direct hardware interface. Spectrometer and probe data are produced by the instruments' own software and reach the system only as files (OE-5, CON-11).

2.8.4 Communications Interfaces

CI-1: The browser and the local application server shall communicate over HTTP and WebSocket on the loopback interface only (default port 8501).

CI-2: The system shall not send e-mail, messages or any network traffic outside the local computer (SEC-5).

Chapter 3: Design and Architecture

This chapter gives a short architecture overview. Its purpose in this SRS is to show how the chosen structure supports the non-functional requirements of Section 2.7; detailed design models will follow in the design phase.

3.1 System Architecture Overview

3.1.1 Purpose

Figure 3.1 shows the conceptual architecture. The whole system runs on the user's computer. The browser talks to a local Streamlit server over HTTP on the loopback interface. Requests pass from the presentation layer to an application-services layer, which validates inputs, orchestrates and caches simulations, manages sessions, parses imported files, runs validation, trains and registers AI models and builds exports. The services call the domain layer, which holds the two core modules and has no user-interface code: Core Module 1, the Digital Twin, and Core Module 2, the AI Module. The AI module learns from data generated by the twin and calls the twin to verify its recommendations; the twin never depends on the AI. The data layer holds the SQLite database, the read-only reference data files, the imported measured files, the trained AI models with their MLflow records, and the exported outputs.

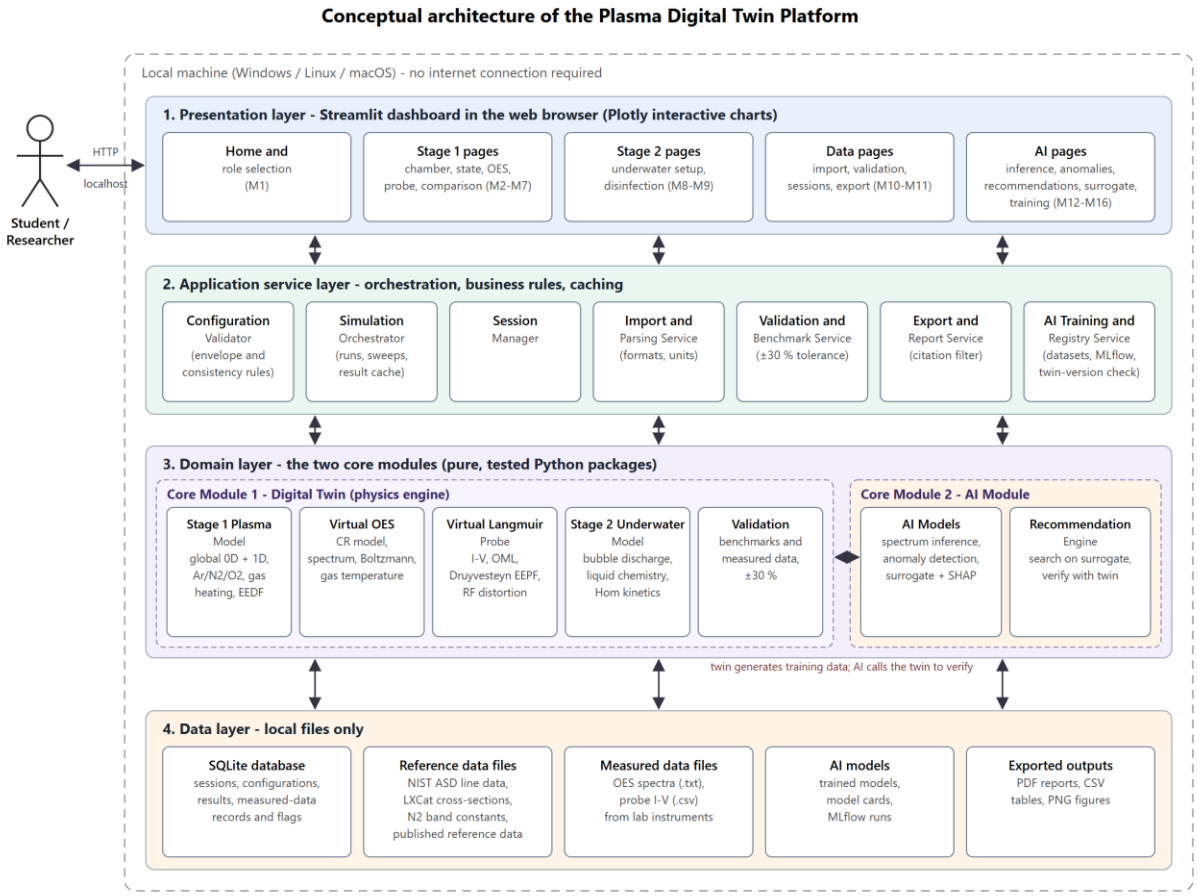


Figure 3.1: Conceptual architecture of the Plasma Digital Twin Platform

3.1.2 Technologies and Services

Table 3.1 lists the components, their technologies and their security mechanisms.

Table 3.1: Technologies, components and security mechanisms used in the system architecture

Component	Description	Technology used	Security mechanism
Presentation layer	Screens M1–M16, navigation, charts	Streamlit, Plotly	Bound to 127.0.0.1; input widgets restrict type and range
Configuration Validator	Envelope and consistency checks (E3, E5)	Python, shared range file	Rejects out-of-range inputs before any computation
Simulation Orchestrator	Runs, caches and checks models; sweeps (E2, E4, E17, E20)	Python, Streamlit cache	Accepts results only after conservation checks
Session Manager	Save, list, open, compare, delete sessions	SQLite (sqlite3)	Transactions; parameterised queries
Import & Parsing	Reads measured .txt and .csv files (E13, E14)	pandas	Type, size and schema checks; data never executed
Validation & Benchmark	±30 % checks against literature and measured data (E15, E16)	Python	Tolerance and references read from versioned file

Component	Description	Technology used	Security mechanism
Export & Report	PDF, CSV and PNG exports (E19)	Plotly static export, pandas	Citation filter removes non-citable data
AI Training & Registry	Twin datasets, training, evaluation, model cards, twin-version check (E22, E23)	scikit-learn, XGBoost, MAPIE, MLflow	Dataset settings locked before training; models never trained on measured data
Core Module 1: Digital Twin	Stage 1 model, EEDF/rates, virtual OES, virtual probe, Stage 2 model, validation	NumPy, SciPy	No file or network access; pure functions
Core Module 2: AI Module	Inference, anomaly detection, surrogate with SHAP, recommendation engine	scikit-learn, XGBoost, SHAP, SciPy	Read-only access to the twin; outputs labelled as AI
Data layer	SQLite database, reference data, measured files, outputs	SQLite, CSV/text files	Checksums for reference data; measured files read-only

3.2 Architecture Style and Support for Non-Functional Requirements

The system uses a **layered architecture** with four layers (presentation, application services, domain and data), in which each layer calls only the layer below it. This style was chosen because it keeps the physics code free of user-interface code, so each model can be tested alone and explained alone; because a desktop tool for one user does not need the deployment complexity of client–server or microservice styles; and because the same domain layer can later serve another front end without change. Inside the domain layer, the dependency between the two core modules runs one way only: the AI module calls the twin, but the twin never imports the AI code, so the physics can always be tested and defended on its own. Table 3.2 shows how the architecture supports the non-functional requirements.

Table 3.2: Architectural support for the non-functional requirements

Quality attribute	Requirements	Architectural decision
Reliability	REL-1 to REL-4	Central error handling in the services layer maps every exception to an event response; SQLite transactions protect sessions; seeds and model versions are stored.
Usability	USE-1 to USE-5	One configuration file drives ranges, units, messages and validation in every screen; shared chart and status components.
Performance	PER-1 to PER-10	Simulation results are cached by configuration hash; the instrument model is separate from the plasma model so it can be reapplied without re-solving; vectorised numerical code.
Security	SEC-1 to SEC-5	Loopback-only server; import and export pass through single services that apply type checks and the citation filter; parameterised data access.
Accuracy	ACC-1 to ACC-4	Domain layer isolated and fully unit-tested; conservation checks before acceptance; versioned reference data and tolerance.
AI model quality	AIQ-1 to AIQ-6	One-way dependency from AI to twin; every model stamped with the twin version and logged in the registry; the recommendation engine cannot return a result without a twin call.

Quality attribute	Requirements	Architectural decision
Maintainability, portability	MNT-1 to MNT-3, POR-1	One module per physical model with its own test file; constants in data files; pure-Python dependencies pinned in one requirements file.

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Appendix A: Interactive Mockup Prototype

The mockups in Section 2.6 were produced as a single clickable HTML prototype so that the supervisor and evaluators can walk through the workflow in any web browser without installing anything.

Prototype file submitted with this SRS: *mockups/interactive_mockups.html*. Use case and architecture diagrams: *mockups/diagrams.html*.

A.1 How to Use the Prototype

- Open the file in Chrome, Edge or Firefox. The left panel lists all screens grouped by stage, each with its mockup ID.
- Buttons such as Run simulation, Send to virtual OES and Compare diagnostics navigate along the real workflow.
- The Student / Researcher switch at the top right shows or hides Researcher-only elements (import, overlays, validation).
- A screen can be opened directly by adding its ID to the address, for example interactive_mockups.html#M5.

A.2 Screen Index

Table A.1 lists every screen in the prototype with its figure and requirement range.

Table A.1: Screens of the interactive prototype

ID	Screen	Stage / group	Figure	Requirements
M1	Home Screen	General	Figure 2.2	FR1.1–FR1.6
M2	Low-Pressure Chamber Configuration	Stage 1	Figure 2.3	FR2.1–FR2.13
M3	Low-Pressure Plasma State Results	Stage 1	Figure 2.4	FR3.1–FR3.10
M4	Virtual OES – Synthetic Spectrum	Stage 1	Figure 2.5	FR4.1–FR4.8
M5	OES Analysis – Boltzmann Plot and Temperatures	Stage 1	Figure 2.6	FR5.1–FR5.10
M6	Virtual Langmuir Probe	Stage 3 (on Stage 1 plasma)	Figure 2.7	FR6.1–FR6.10
M7	Diagnostics Comparison	Stage 1 / 3	Figure 2.8	FR7.1–FR7.6
M8	Underwater Plasma Configuration	Stage 2	Figure 2.9	FR8.1–FR8.13
M9	Underwater Plasma and Disinfection Results	Stage 2	Figure 2.10	FR9.1–FR9.6
M10	Measured Data Import and Validation	Data (Researcher)	Figure 2.11	FR10.1–FR10.8
M11	Session History and Export	Data	Figure 2.12	FR11.1–FR11.5
M12	AI Parameter Inference from Spectra	AI Module	Figure 2.13	FR12.1–FR12.6
M13	AI Physics-Relationship Anomaly Detection	AI Module	Figure 2.14	FR13.1–FR13.7
M14	AI Recommendations (Inverse Design)	AI Module	Figure 2.15	FR14.1–FR14.8
M15	AI Surrogate Model and Explainability	AI Module	Figure 2.16	FR15.1–FR15.6

ID	Screen	Stage / group	Figure	Requirements
M16	AI Training, Evaluation and Model Registry	AI Module	Figure 2.17	FR16.1–FR16.5

Appendix B: Requirements Traceability Matrix

Table B.1 traces each business objective to the modules, use cases, mockups, functional requirements and non-functional requirements that deliver it, so that any feature can be followed from objective to requirement.

Table B.1: Requirements traceability matrix

Objective	Sub-modules	Use cases	Mockups	Functional requirements	NFRs
BO-1 Stage 1 twin within ±30 %	1.1, S.1	UC-01–UC-05	M1–M3	FR1.2, FR2.1–FR2.13, FR3.1–FR3.10, FR17.1–FR17.4, FR17.6, FR17.14	PER-1, PER-4, REL-1, ACC-1, ACC-3
BO-2 OES and probe in < 5 min	1.2, 1.3, S.1, S.2	UC-06–UC-09, UC-11–UC-13	M1, M4–M6	FR1.1, FR1.4–FR1.6, FR4.1–FR4.8, FR5.1–FR5.10, FR6.1–FR6.10, FR17.7–FR17.11	PER-2, PER-3, USE-1–USE-5
BO-3 Diagnostic consistency	1.2, 1.3	UC-14	M7	FR7.1–FR7.6, FR5.5	ACC-2
BO-4 Disinfection within ±30 %	1.4	UC-15–UC-19	M8, M9	FR8.1–FR8.13, FR9.1–FR9.6, FR17.5, FR17.17	PER-5, ACC-1
BO-5 Validation report in < 10 min	1.5, S.1, S.2	UC-20, UC-21, UC-22, UC-23	M1, M10, M11	FR1.3, FR10.1–FR10.8, FR11.1–FR11.5, FR17.12, FR17.13, FR17.15, FR17.16	PER-7, SEC-2, SEC-3, ACC-4
BO-6 AI spectrum inference	2.1, 2.5	UC-10, UC-28	M12, M16	FR12.1–FR12.6, FR16.1–FR16.5, FR17.18, FR17.19	PER-9, AIQ-1, AIQ-5, AIQ-6
BO-7 Physics anomaly detection	2.2	UC-24	M13	FR13.1–FR13.7, FR17.20	AIQ-2
BO-8 Surrogate with SHAP	2.3	UC-27	M15	FR15.1–FR15.6, FR17.22	AIQ-3
BO-9 Twin-verified recommendations	2.4	UC-25, UC-26	M14	FR14.1–FR14.8, FR17.21	PER-10, AIQ-4

The remaining non-functional requirements (REL-2 to REL-4, PER-6, PER-8, SEC-1, SEC-4, SEC-5, MNT-1 to MNT-3 and POR-1) are cross-cutting and apply to every objective.

Appendix C: Resolved Parameter Values and Sources

The domain values used in the business rules were collected from the supervisor's written answers to two questionnaire rounds. Where the supervisor left an item blank, a published source was used, and where an answer needed interpretation, the design decision is stated. Table C.1 lists each value and its source so that every business rule can be defended.

Table C.1: Resolved parameter values and their sources

Item	Value used in this SRS	Source
Stage 1 discharge	13.56 MHz CCP; 10–100 W; 0.1–0.5 mbar (75–375 mTorr)	Supervisor
Stage 1 chamber	40 × 40 cm; electrodes 14 cm, gap 4.5 cm	Supervisor (size); COMSATS laboratory paper [15] (electrodes)
Stage 1 gases	Ar 100–500 sccm; N_2 and O_2 each 5 % of the Ar flow	Supervisor
Gas mixing scheme	Four selectable modes (Ar, Ar+N_2, Ar+O_2, Ar+N_2+O_2)	Design decision (answer did not state whether gases are added together)
Stage 1 EEDF	Druyvesteyn (Maxwellian selectable)	Supervisor
Stage 1 n_e for OES	Langmuir probe value; model value until a measured probe value is imported	Supervisor; interpretation to avoid circularity
Gas temperature	N_2 (1,0) band at 315.9 nm with NH check; Doppler when resolution allows	Supervisor; [13]
Spectrometer	High resolution, better than 0.01 nm (default 0.005 nm)	Supervisor
Langmuir probe	Single cylindrical tungsten, 0.195 mm × 5 mm, −20 to +50 V, 0.5 V steps; OML; RF distortion	Supervisor (OML, RF); [15] (dimensions, sweep)
Stage 2 discharge	Pulsed DC, up to 10 kV, 5–10 kHz, 100 W, pin-to-pin 1/2/3 mm, 1 atm	Supervisor
Stage 2 pulse width	Configurable in µs with duty cycle < 1 (default 10 µs)	Design decision (100 µs at 10 kHz gives 100 % duty)
Stage 2 gas	O_2 100–300 sccm (default 100 sccm) with water vapour	Supervisor; water vapour [17]
Stage 2 EEDF and n_e	Maxwellian; H-alpha Stark broadening	Supervisor; [14]
Waste water	Municipal, medium strength: COD 430 mg/L, pH 7.5, 25 °C, ≈ 750 µS/cm	Supervisor (municipal); [19] (values)
Organisms and dose	*E. coli* (also *S. aureus*, bacterial spores); 1–2 × 10^5 CFU/mL; 1 min	Supervisor
Disinfection kinetics	Hom model, parameters from published time-resolved data	Supervisor; [7]
Tolerance and 1D	±30 %; 1D only for Stage 1 profiles	Supervisor (accepted default)
Measured data	Validation only; not citable; probe .csv, OES text	Supervisor
AI module	Core Module 2: inference, anomaly detection, surrogate with SHAP, recommendations	Team decision, 29-Oct-2026 (the supervisor's Round 3 answer had been "optional, decide later")

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